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Telecom & Cable Services, GVAT

The Telecom Investors' Accounting Handbook

We dive deep into accounting and capital structure complexities at T, VZ and TMUS that impact investors' views of earnings, FCF, enterprise value, leverage and ultimately valuation, and extend our analysis to US Cable and Canadian Telecoms where similar dynamics exist. We reiterate AT&T as Top Pick.



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For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

Contributors



MORGAN STANLEY & CO. LLC
Benjamin Swinburne, CFA
Equity Analyst
+1 212 761-7527
Benjamin.Swinburne@morganstanley.com



MORGAN STANLEY & CO. LLC
Lindsay A Tyler
Credit Analyst
+1 212 761-2734
Lindsay.Tyler@morganstanley.com



MORGAN STANLEY & CO. LLC
Kyle B Yellin
Credit Analyst
+1 212 761-4706
Kyle.Yellin@morganstanley.com



MORGAN STANLEY & CO. LLC
Mariah Thompson
GVAT Strategist
+1 212 761-1147
Mariah.Thompson@morganstanley.com



MORGAN STANLEY & CO. LLC
Todd Castagno, CFA, CPA
GVAT Strategist
+1 212 761-6893
Todd.Castagno@morganstanley.com



MORGAN STANLEY & CO. LLC
Blake Netter, CPA
Research Associate
+1 212 761-4679
Blake.Netter@morganstanley.com



MORGAN STANLEY & CO. LLC
Patrick A Ho
Research Associate
+1 212 761-2764
Patrick.Ho@morganstanley.com



MORGAN STANLEY & CO. LLC
Kate Konetzke, CFA, CPA
GVAT Strategist
+1 212 761-3457
Kate.Konetzke@morganstanley.com



MORGAN STANLEY & CO. LLC
David M Hamburger
Credit Analyst
+1 212 761-1580
David.Hamburger@morganstanley.com



MORGAN STANLEY & CO. LLC
Clinton Chang, CFA, CPA
GVAT Strategist
+1 212 761-1185
Clinton.Chang@morganstanley.com

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The Telecom Investors' Accounting Handbook

Key Takeaways

- Comparables matter—we normalize earnings, FCF, enterprise value, and debt across T, VZ, and TMUS, pushing valuation and leverage multiples higher.
 - The accounting and tax treatment of wireless spectrum are a gift to the US carriers. Wireless EPS and FCF multiples are more flattering vs. Cable as a result.
 - We believe AT&T and Verizon are best-positioned to benefit from domestic tax policy, which would accelerate FCF and capital return to shareholders.
 - We analyze promo bill credit trends, which are rising albeit at a moderating pace, and examine industry EIP securitization programs and their financial impacts.
 - Wireless operators carry large debt levels and manage to rating agency debt terms, yielding adjusted leverage ~0.6x higher than reported.
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The US Telecom industry has consolidated significantly with three primary operators - AT&T, Verizon and T-Mobile - and Comcast and Charter more recently entering the market. All five carry large balance sheets, invest substantial capital in their businesses, and operate through complex customer contracts. This has led to increasingly complex accounting that is often not applied consistently across the industry.

We combine the expertise of our Equity, Global Valuation, Accounting & Tax (GVAT), and Credit teams for a comprehensive perspective on the accounting, capital structure, tax, and valuation issues we believe are most important when exploring investment opportunities in the sector. Where possible, we present adjustments

that improve comparability across companies to more accurately reflect the underlying economics of transactions. For complex or less quantifiable trends, we identify metrics to track and highlight critical areas of risk or potential upside.

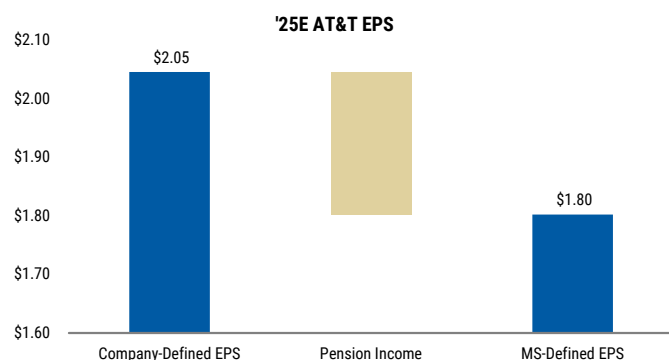
As a companion to this report, see our 2025 Year-Ahead Outlook "Bringing It All Together" for our investment views [here](#). On the equity side, we remain OW T (Top Pick), TMUS, EW CMCSA, CHTR, SATS, T.TO and ATUS, and UW BCE, RCIb, and SIRI. On the credit side, we recommend buying select AT&T and TMUS bonds, see opportunities in high yield Cable & Satellite, and are more cautious on Canadian Telecoms.

Executive Summary: In Telecom, Accounting Plays A Key Role in Investment Analysis

Normalizing Earnings & FCF Across the Wireless Group

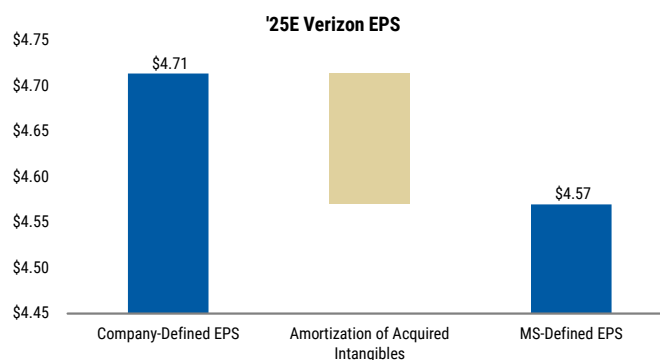
Our approach to valuation in the Telecom & Cable Services sector is to attempt to normalize earnings and FCF across the group, fully burdening earnings and cash flows for all recurring operating and financing costs, cash and "non-cash." We also attempt to exclude non-operating, non-recurring transactions. The charts below outline the adjustments we make to the financials of each of the major operators.

Exhibit 1: We adjust AT&T EPS to exclude non-operating, non-cash pension income, which increases our 2025 P/E multiple by ~1.8x to 15.1x compared to company methodology of 13.3x.



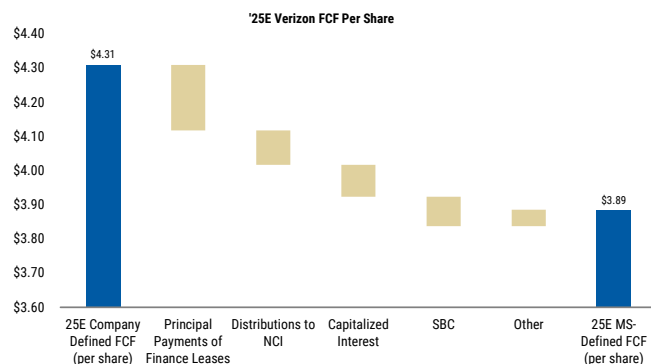
Source: Company Filings and Morgan Stanley Research

Exhibit 2: We adjust Verizon EPS to include amortization of acquired intangible assets. This adjustment does not have a material impact on our 2025 P/E.



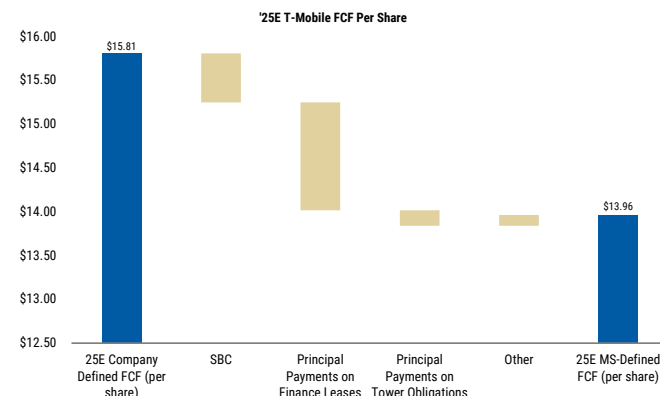
Source: Company Filings and Morgan Stanley Research

Exhibit 3: We adjust Verizon FCF to include principal payments on finance leases, distributions to NCI, interest capitalized to spectrum, and SBC. This increases our 2025 P/FCF by ~1x turn to 11.1x compared to 10x under company methodology.



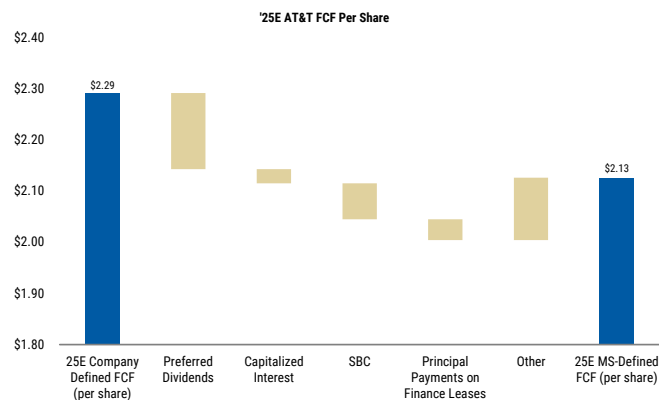
Source: Company Filings and Morgan Stanley Research

Exhibit 4: We adjust T-Mobile FCF to include SBC and principal payments on finance leases and tower obligations. These adjustments increase our 2025 P/FCF by ~2x turns (16x under company methodology, to 18x after adjustments, and 23x fully taxed).



Source: Company Filings and Morgan Stanley Research

Exhibit 5: We adjust AT&T FCF for preferred dividends, interest capitalized to spectrum, SBC, and principal payments on finance leases. These adjustments increase our 2025E P/FCF by ~1x turn to 12.8x vs 11.8x under company methodology.



Source: Company Filings and Morgan Stanley Research

Exhibit 7: T: We include minority interest and tax-adjusted unfunded pension and other postretirement benefit obligations in EV. These adjustments increase our EV/EBITDA multiple by 0.6x to 7.2x.

	Current Price		
	2024	2025E	2026E
Shares Outstanding (FD)	7,181.00	7,112.62	6,874.40
Share Price	\$27.16	\$27.16	\$27.16
Total Market Value	\$195,036.0	\$193,178.7	\$186,708.6
Plus:			
Total Debt	\$123,532.0	\$119,873.4	\$114,538.9
Preferred Stock	\$20,280.0	\$20,280.0	\$20,280.0
Other Obligations (Pension/OPEB)	\$7,002.6	\$7,241.2	\$7,489.6
Less:			
Cash and Cash Equivalents	\$3,298.0	\$8,592.6	\$3,892.0
Enterprise Value	\$342,552.6	\$331,980.8	\$325,125.1
Total EBITDA	\$44,759.2	\$46,376.7	\$47,978.0
EV / EBITDA	7.7x	7.2x	6.8x
Forward Total EBITDA	\$46,376.7	\$47,978.0	\$50,195.3
EV / Fwd. EBITDA	7.4x	6.9x	6.5x

Source: Company Filings and Morgan Stanley Research

Exhibit 6: VZ: We include minority interest and tax-adjusted unfunded pension and other postretirement benefit obligations in EV. These adjustments do not have a material impact on our EV/EBITDA calculation.

	Current Price		
	2024	2025E	2026E
(\$ Millions, except per-share data)			
Fully Diluted Shares Outstanding	4,214.70	4,223.62	4,226.04
Share Price	\$43.19	\$43.19	\$43.19
Total Market Value	\$182,032.9	\$182,418.0	\$182,522.7
Plus:			
Debt	\$144,014.0	\$140,280.3	\$137,251.5
Other Liabilities	0.0	0.0	0.0
Total Debt	\$144,014.0	\$140,280.3	\$137,251.5
Preferred Stock	\$0.0	\$0.0	\$0.0
Minority Interest	\$1,338.0	\$1,338.0	\$1,338.0
Other Obligations (Pension/OPEB)	\$7,747.6	\$6,913.2	\$5,940.4
Less:			
Cash and Cash Equivalents	\$4,194.0	\$4,405.3	\$6,397.0
Enterprise Value	\$330,938.5	\$326,544.2	\$320,655.7
Total EBITDA	\$48,791.0	\$50,341.9	\$51,886.5
EV / EBITDA	6.8x	6.5x	6.2x
Forward Total EBITDA	\$50,341.9	\$51,886.5	\$52,987.7
EV / Fwd. EBITDA	6.6x	6.3x	6.1x

Source: Company Filings and Morgan Stanley Research

Exhibit 8: TMUS: We include the value of T-Mobile's unconsolidated fiber JV's in EV. This adjustment does not have a material impact on our EV/EBITDA multiple.

	Current Price		
	2024	2025E	2026E
Fully Diluted EOP Shares Outstanding	1,148.0	1,115.5	1,059.8
Share Price	\$262	\$262	\$262
Total Market Value	\$300,992.6	\$292,451.3	\$277,851.1
Plus:			
Debt	\$80,591.0	\$82,857.2	\$93,081.2
Other Liabilities	0.0	0.0	0.0
Total Debt	\$80,591.0	\$82,857.2	\$93,081.2
Less:			
Cash and Cash Equivalents	\$5,409.0	\$2,696.0	\$2,696.0
Unconsolidated Assets			
MetroNet (50%)	\$4,900.0	\$4,900.0	\$4,900.0
Lumos (50%)	950.0	950.0	950.0
Total Unconsolidated Asset Value	\$5,850.0	\$5,850.0	\$5,850.0
Enterprise Value	\$370,324.6	\$366,762.6	\$362,386.3
Total EBITDA	\$31,278	\$33,018	\$35,541
EV / EBITDA	11.8x	11.1x	10.2x
Forward Total EBITDA	\$33,018	\$35,541	\$38,390
EV / Fwd. EBITDA	11.2x	10.3x	9.4x

Source: Company Filings and Morgan Stanley Research

Exhibit 9: VZ: Verizon's use of ABS debt versus off-balance-sheet installment plan receivables sales used by peers makes the EV/EBITDA multiple not apples-to-apples. Excluding ABS debt from Verizon's EV would reduce its current EV/EBITDA trading multiple by approximately half a turn.

	Current Price		
(\$ Millions, except per-share data)	2024	2025E	2026E
Fully Diluted Shares Outstanding	4,214.70	4,223.62	4,226.04
Share Price	\$43.19	\$43.19	\$43.19
Total Market Value	\$182,032.9	\$182,418.0	\$182,522.7
<i>Plus:</i>			
Debt	\$144,014.0	\$140,280.3	\$137,251.5
Other Liabilities	0.0	0.0	0.0
Total Debt	\$144,014.0	\$140,280.3	\$137,251.5
Preferred Stock	\$0.0	\$0.0	\$0.0
Minority Interest	\$1,338.0	\$1,338.0	\$1,338.0
Other Obligations (Pension/OPEB)	\$7,747.6	\$6,913.2	\$5,940.4
<i>Less:</i>			
Cash and Cash Equivalents	\$4,194.0	\$4,405.3	\$6,397.0
Enterprise Value	\$330,938.5	\$326,544.2	\$320,655.7
Less: Asset-Backed Debt	\$26,337.0	\$26,337.0	\$26,337.0
Enterprise Value less Asset-Backed Debt	\$304,601.5	\$300,207.2	\$294,318.7
Total EBITDA	\$48,791.0	\$50,341.9	\$51,886.5
EV / EBITDA	6.8x	6.5x	6.2x
Total EBITDA	\$48,791.0	\$50,341.9	\$51,886.5
EV less Asset-Backed Debt / EBITDA	6.2x	6.0x	5.7x
Forward Total EBITDA	\$50,341.9	\$51,886.5	\$52,987.7
EV / Fwd. EBITDA	6.6x	6.3x	6.1x
Forward Total EBITDA	\$50,341.9	\$51,886.5	\$52,987.7
EV less Asset-Backed Debt / Fwd. EBITDA	6.0x	5.8x	5.6x

Source: Company Filings and Morgan Stanley Research

Summary of Takeaways Across Accounting Topics

In this in-depth report, we describe and analyze how the major wireless operators account for a wide range of operating activities and financing decisions. We summarize and then analyze the key topics that we believe investors in the space should understand. These accounting approaches are particularly important to understand when forecasting earnings power, free cash flow, debt leverage, return on invested capital, and other important investor tools. The fact that wireless operators across the industry use a range of approaches when adopting these accounting guidelines further highlights the importance of this work.

In our equity and credit investment work, we also make adjustments to the accounting discussed below in order to 1) reduce the inconsistencies in reported financials across the sector and 2) arrive at what we believe are the most appropriate measures of earnings, assets, liabilities, and cash flow.

Key Accounting Topics for US Wireless Companies

TOPIC (WITH SECTION LINK)	KEY POINTS
Equipment Installment Payment Plans: Revenue Recognition	With EIP plans, the timing of revenue recognition and operating cash flows are mismatched: Wireless carriers generate revenue from bundled offerings, often bundling equipment, services, and payment plans in one agreement. Revenue recognition accounting requires that the transaction price is allocated to each performance obligation in the contract, with discounts recognized over time. Installment billing results in revenue recognition ahead of cash collections, creating a disconnect between revenue recognition and cash flows. We estimate Verizon's service revenue growth is negatively impacted by promotional discounts on phone upgrades through 2025.
Equipment Installment Plan Receivables	With EIP financing, operators can finance handset subsidies, front-load cash flows: To accelerate cash collections, wireless carriers engage in receivables factoring and securitization. Each carrier has its own strategy for monetizing receivables, resulting in different cash flow and balance sheet profiles. Securitizations can follow what happens with device sales, so lower upgrade rates often pair with less securitization growth. TMUS, T, CHTR, and VZ all use different approaches to EIP financing which lead to different implications for metrics including leverage, ROIC, EV, cash collection cycle, and days sales outstanding.
Supplier Financing Programs	Vendor/supplier financing programs can create lumpiness in reported cash flow and shift operating cash flows to financing cash flows: AT&T and Verizon have supplier financing programs to manage working capital by extending the payment terms on their payables. These programs offer attractive financing terms and extend payables, but create lumpiness to reported cash flows, and shift some cash flows from Cash Flows From Operations on statement of cash flows to Cash Flows From Financing. When we forecast FCF for valuation purposes, we include cash flows reported in Cash Flows from Financing as operating cash flows with respect to supplier and vendor financing. Many operators now follow this treatment, including all major US operators. We also monitor supplier financing activity in Cash Flow from Operations as these programs can have a significant impact on reported cash flow.
Spectrum Licenses - The Accounting & Tax Gift to Wireless Carriers	The accounting and tax treatment of spectrum investments is highly beneficial to wireless operators: The Big 3 US wireless carriers have invested billions in spectrum licenses to support and expand their networks. These assets are not amortized under accounting rules, and investments are reported separately from other capital expenditures. As a result, spectrum licenses, a core operating asset, are not factored into either EPS or FCF. Spectrum accounting favors wireless assets relative to wireline assets, leaving trading multiples for wireless carriers arguably understated compared to Comcast and Charter, as examples. We include spectrum costs in our DCF valuation analysis.

Finance Leases

Principal payments for finance leases should be viewed as operating cash flows, although they are reported in cash flows from financing: Network infrastructure can be purchased or leased. Purchases and operating leases are captured in free cash flow, but principal payments on finance leases are considered financing cash flows. T-Mobile uses finance leases more than its two peers. We adjust FCF in our valuation work to capture principal payments for leases. We also cover the accounting for tower obligations resulting from failed sale leaseback transactions.

The most significant impact to FCF estimates from incorporating principal payments for finance leases is in the three Canadian operators, BCE, RCiB, and Telus.

Pensions and Other Post-Employment Benefits

Underfunded pension and other post-employment obligations should be included as debt when building enterprise value: AT&T and Verizon employ a large number of unionized employees who are entitled to pension and other retirement benefits. Both companies have improved the funding of their pension obligations and transferred some of the risk to third parties in recent years; however, because the obligations are so large, they can still have outsized impacts on earnings.

AT&T is most impacted by our pension related adjustments, including reducing earnings for pension-related income and increasing enterprise value for underfunded pension obligations.

Taxes

Because wireless carriers are capital intensive, they are some of the biggest beneficiaries of bonus depreciation tax policy. In addition, sophisticated tax strategies are used to further reduce cash taxes. We cover the economic benefit of bonus depreciation and the accounting for uncertain tax positions.

New bonus depreciation legislation would be a clear benefit to FCF across Telecom and Cable. However, AT&T, Verizon, and EchoStar have meaningful uncertain tax benefits relative to their size which are in fact cash tax savings taken in past periods. These disclosures highlight the potential cash tax liability should their historical deductions be reversed by the IRS. Comcast, T-Mobile, and Charter also have uncertain tax benefits.

Amortization of Acquired Intangibles

Amortization of intangibles can be both operating and/or acquisition related: The most common operating activity that creates amortization of intangibles is capitalized software, which we see in Verizon's intangibles. Acquisitions also create intangibles subject to amortization (and goodwill, which is not amortized). Many operators add back the amortization of acquired intangibles; however, these acquisitions are generating earnings. We believe it is more consistent to include both the earnings and the amortization of the cost of acquiring these earnings when assessing earnings power.

We include the amortization of acquired intangibles in Verizon's earnings. We exclude the amortization of DirecTV related acquired intangibles in estimating AT&T's earnings, as AT&T expects the sale of its stake to close in 2H25.

Preferred Interests and Wireless Partnerships

The wireless business has consolidated over the decades, but some partnerships remain in place: These partnerships are typically carried as non-controlling interest or preferred equity. These are earnings not attributable to common shareholders, typically carry cash distribution obligations, and may have put/call features. We include these NCI's and/or pfd's as liabilities in building up enterprise value and we include the expense and/or distributions to partners as reductions to earnings or FCF.

AT&T has the largest preferred interests and the largest adjustments in our models.

Adjusting Reported Debt for Debt-Like Obligations

For the related obligations of many of the key accounting topics we discuss, credit investors and ratings agencies make adjustments to respective debt balances (e.g., lease liabilities, EIP programs, pensions). For this broader audience across equity and credit investors, we think it is worth noting the size of such adjustments to debt and leverage, the differing methodologies for different obligations, key read-throughs for corporate funding approaches and changes since the C-band auction, and the presence of structurally senior obligations. We outline these key takeaways here, with more detail shared in a later section: [Debt Adjustments by Rating Agencies & Credit Investors](#).

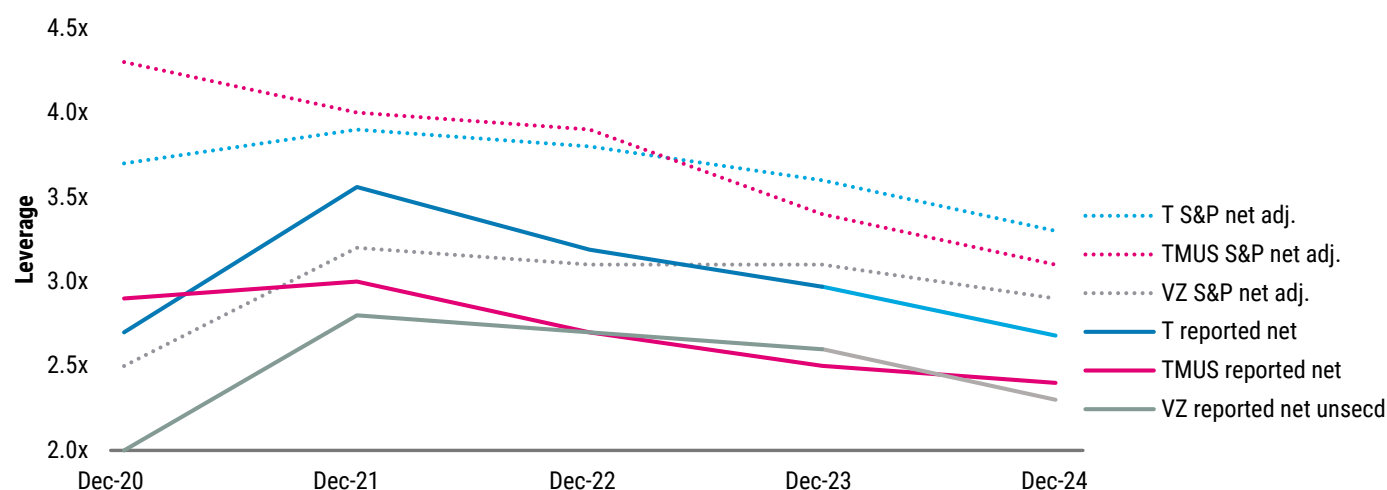
Four Key Learnings Related to Debt Adjustments for US Wireless Companies	
KEY TAKEAWAY	DETAILS
Sizable Adjustments to Debt and Leverage	These adjustments to reported debt are sizable, shown here in Exhibit 10. We estimate more than \$35bn of adjustments for each of the Big 3 wireless companies, accounting for ~30% of gross adj. debt (when counting ABS as an adjustment). Those metrics are markedly higher than what we see for the rest of our broader investment grade TMT fixed income coverage. These debt adjustments overrun agencies' adjustments to EBITDA and other metrics, leaving adjusted leverage significantly higher (as much as 0.7x higher) vs. reported leverage. See Exhibit 11.
Different Methodologies for Adjustments	Methodologies can differ across rating agencies and credit investors, yielding different results for adj. debt balances. A straightforward example is that S&P nets out cash, while Moody's does not. A more complex example is in regards to handset equipment plans. Moody's views handset ABS and EIP securitization cash proceeds as debt (which we agree with), S&P gives a captive finco benefit (though treatment differs between ABS & private securitizations), and credit investors have different opinions on the topic.
Insights on Big 3 Funding Approaches & Changes	Comparing the US wireless companies' reported and adjusted debt balances can illuminate the different approaches management teams have taken, changes in strategies, and impacts of corporate moves. There is also value in comparing those balances over time as well, specifically from YE20 before the C-band auction, vs. YE24 (see Exhibit 60). At Verizon, gross adj. debt has almost fully recovered, ending 2024 \$15-20bn wide of its YE20 mark, which is largely explained by the increase in additional secured ABS debt. However, the company announced this fall an agreement to acquire Frontier (FYBR) in a \$20bn all-cash deal, which is pending. At TMUS, we've seen limited changes over the approx. four-year lookback, but expect balance sheet debt and handset ABS to grow in the future. AT&T stands out for not only for its debt reduction after the C-band auction via corporate moves, but also its generally more diversified funding approach over the years (preferred shares, preferred subsidiary interests, private EIP program etc.). AT&T's corporate moves following the auction, namely the stake sale of DIRECTV to TPG (Jul'21) and AT&T's spin-merge of Time Warner (Apr'22), resulted in gross adj. debt moving from a multi-year-sustained \$210bn level down to the \$180bn area now, via reduction of both unsecured debt and debt-like obligations.
Presence of Structurally Senior Obligations	Some obligations can be considered as structurally senior to unsecured debt at the parent (i.e., obligations such as pension underfunding, preferred equity at subsidiaries, securitization, opco debt, proportionate consolidation for debt issued by a partly-owned asset). AT&T has the most notable exposure. However, structurally senior obligations are arguably capped. Maintaining an appropriate ratio of structurally senior debt to total debt, typically 20%, is required to maintain a typical IG structure with no ratings notching between layers of debt. This requirement gives us piece of mind that debt-like obligations, and with that debt adjustments, will not be overused.

Exhibit 10: The Big 3 US wireless names require numerous debt adjustments on top of their reported debt metric, including but not limited to: pension, lease liabilities, hybrid instruments, preferred equity at subsidiaries

YE24 Debt Adjustments (\$bn)	AT&T	TMUS	VZ	Quick Notes on Context, Agency Treatment, Takeaways
Debt	123.5	76.7	117.9	We exclude on-balance-sheet ABS to start
(+) Handset ABS	0.0	1.6	26.1	Public ABS included in bal. sheet debt, called "secured" by VZ; note S&P gives captive finco benefit
(+) Tower Obligations	0.0	3.7	0.0	Tower obligations line item on balance sheet for TMUS considered debt-like
Approx. Starting Debt	123.5	81.9	144.0	
(+) Other Securitizations	8.2	1.5	0.0	Cash proceeds received for off-balance-sheet EIP program; S&P differs & gives captive finco benefit
(+) Pension	3.0	0.9	1.6	Regulatory and contractual in nature
(+) Operating Leases	20.9	32.0	24.3	Agencies generally incorporate both operating & financing leases, sizable in this industry
(+) Preferred Equity at Parent	2.6	0.0	0.0	Giving 50% equity credit treatment to AT&T's cumulative perpetual preferred shares across Series A, B, C
(+) Preferred Equity at Subsidiaries	15.3	0.0	0.0	T's pfd interests in subs Tower Holdings, Telco LLC, Mobility II (100%); structurally sr to AT&T Inc debt
Subtotal of Other Adjustments	50.0	34.4	25.9	
Adj. Gross Debt (Est. for Moody's)	173.6	116.3	170.0	Showing Moody's approach to gross adj. debt, applying prior published adjustments to YE24 co. reported #s
(-) Cash	(3.3)	(5.4)	(4.2)	Moody's does not net out cash, but S&P does
Adj. Net Debt	170.3	110.9	165.8	
All Adjustments	50.0	39.6	52.1	Subtotal above plus earlier handset ABS & tower ob. line items
Adj. Gross Debt	173.6	116.3	170.0	Shown above, estimate for Moody's
All Adjustments / Adj. Gross Debt	29%	34%	31%	~30-35% across the board

Source: Company filings, Moody's (Peer Snapshot for AT&T dated 3/27/25; Credit Opinion notes dated 6/5/24 for AT&T, 5/8/24 for TMUS, 9/10/24 for VZ, Cross-Sector Rating Methodology note dated 10/9/24 "Financial Statement Adjustments in the Analysis of Nonfinancial Corporations"), S&P (note dated 5/7/24 "Credit FAQ: Calculating Leverage For Selected U.S. Telecommunications And Cable Companies (2024 Update)"), Morgan Stanley Research. Note: See Exhibit 60 for prior period adjustments, and the following table in that section for full methodology details.

Exhibit 11: The Big 3 US wireless companies see ~0.6-0.7 turns of differential between their reported net leverage metrics and respective S&P adjusted net leverage metrics, reflecting the impact of debt-like obligations for these names



Source: Company filings, Morgan Stanley Credit Research, S&P (notes dated 3/5/25 and 12/6/24 for AT&T, notes dated 9/5/24 and 6/13/24 for VZ, notes dated 3/24/25 and 5/16/24 for TMUS; note dated 4/14/25 and titled "Credit FAQ: Calculating Leverage For Selected U.S. Telecommunications And Cable Companies (2025 Update)"). Note: We present S&P's adjusted leverage metric, which nets out cash, here for comparison because Moody's adjusted metric is gross (total).

Equipment Installment Payment Plans: Revenue Recognition

Wireless carriers often use installment plans to bundle equipment, services, and extended payment terms in the same contract. For accounting purposes, the total transaction price is allocated to each performance obligation, with discounts recognized over time. Verizon has indicated its service revenue growth will be negatively impacted through 2025 by amortization of previous promotional discounts on phone upgrades. In this section, we provide an overview of the accounting for bundled transactions, highlight key differences in accounting policies that impact comparability, and present balance sheet analysis to help investors track promotional activity trends at Verizon and AT&T.

Revenue Recognition Overview

Wireless carriers offer customers the ability to pay for their devices on installment plans over a period of 24-36 months as a way to induce equipment sales and retain customers on monthly wireless service plans. The plans often involve promotional discounts, traded-in devices, and/or guaranteed trade-in values.

Installment plans and trade-in programs gained popularity more than a decade ago. Then, new accounting guidance introduced in 2018 altered the way that these transactions were disclosed and presented in financial statements, requiring the phone discount to be recognized over the duration of the service contract. The guidance contained in ASC 606 follows a five-step process for recognizing revenue. We first outline the five-step process for an equipment installment plan sale, then address unique accounting policies at each wireless carrier, and highlight how the accounting shows up across the financial statements.

1. Identify the contract with the customer: Wireless carriers typically incentivize customers to sign up for their service or upgrade their phones by offering payment plans and discounts on future bills. When a customer signs up to pay for their device under an installment plan, the duration of the equipment and installment contract for accounting purposes is typically equal to the duration of the installment plan. This is because while the customer can legally leave the contract at any time, they would forfeit all future bill credits they received when they purchased the device and need to pay off the remaining balance of the installment plan in order to do so.

2. Identify the performance obligations in the contract: An equipment installment plan sale includes the following performance obligations:

- Mobile device;
- Monthly services (wireless service, device protection, etc.);
- A financing component and;
- May or may not include a trade-in or upgrade guarantee, free content trials, etc.

3. Determine the transaction price: The transaction price includes all consideration the company is entitled to receive from the customer less any consideration payable to the customer. For equipment installment plans, this includes:

- Total monthly device payments
- Total monthly wireless service payments
- Fair value of a traded-in device
- Less any trade-in payments or credits for the old device ("consideration payable to the customer")

4. Allocate the transaction price: The transaction price calculated in step 3 is typically less than the total price a customer would pay if the products and services were not bundled. The discount provided on the bundled transaction is allocated to each performance obligation based on the relative standalone selling price (i.e., the price a customer would pay if each good and service were purchased separately).

While the carriers do not explicitly charge the customer interest in the installment plan, implied interest is calculated for accounting purposes to account for the time value of money when the customer pays for a device more than 12 months after it is received.

5. Recognize revenue: Revenue is recognized as the performance obligations are satisfied. For the performance obligations identified in step 2 above, this means:

- *Mobile device* - when the customer takes possession of the phone
- *Monthly services* - as the carrier provides access to the wireless network
- *Financing component* - Interest is deferred at the time the customer receives the device, and interest income is recognized over the duration of the installment plan

Revenue Recognition Accounting Policy Differences Among Wireless Carriers

Slight differences in contract terms or accounting policy elections can result in different accounting treatment for similar offerings. For equipment installment plans, the wireless carriers have varying accounting policies for bill credits and financing components. Verizon accounts for bill credits differently than T-Mobile and AT&T, resulting in different balance sheet presentation that makes analyzing promotional trends and receivables trends across companies more challenging for investors. T-Mobile accounts for a significant financing component across all installment plans, while Verizon considers the financing component to be insignificant for some of its plans, and AT&T does not disclose its financing component accounting policy. A significant financing component means that some of the transaction price is allocated to interest income instead of the device or wireless services, meaning gross margin may not be as comparable.

Promotional Billing Credit Trend Analysis

AT&T and Verizon apply different accounting treatments for promotional billing credits. AT&T discloses that accounts receivable are recorded *net* of promotional billing credits. At the time of handset sale, AT&T records the discount allocated to monthly service, which will reduce service revenue over the term of the contract, as a *contract asset*. Verizon does not reduce accounts receivable for promotional billing credits and instead records them as a *contract liability* on the balance sheet.

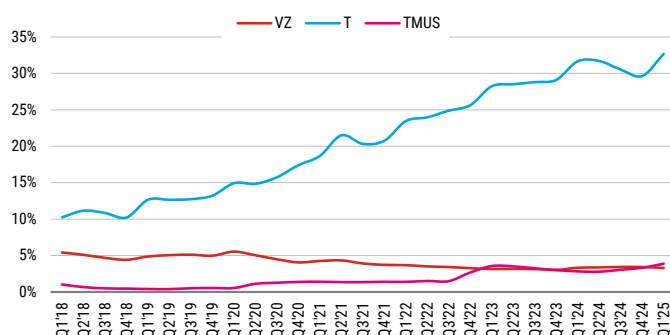
Verizon and AT&T Revenue Recognition Accounting Policy Disclosures

AT&T 2024 10-K: REVENUE RECOGNITION FOOTNOTE

"When installment sales include promotional discounts that are earned by customers over the contract term (e.g., "buy one get one free" or equipment discounts with trade-in of device), notes receivable are recognized net of discounts and the difference between revenue recognized and consideration received is recorded as a contract asset to be amortized over the term"

While these accounting policies do not affect revenue or ARPU, balance sheet metrics will not be comparable. **The accumulation of bill credits on the balance sheet relative to device revenue can provide an indication of promotional activity levels as well as discounts remaining to be amortized through revenue over the duration of the installment plan.** However, because these companies have elected different accounting policies, investors need to adjust their analysis.

Exhibit 12: Rising contract assets at AT&T is an indicator of promotional activity levels



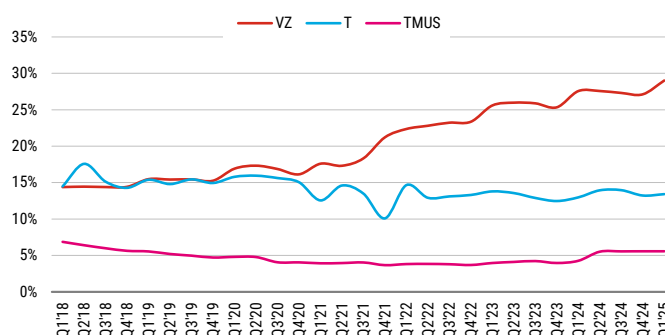
Source: Company filings & Morgan Stanley Research.

VERIZON 2024 10-K: REVENUE RECOGNITION ACCOUNTING POLICY

"For our device payment plans, billing credits are accounted for as consideration payable to a customer and are included in the determination of total transaction price, resulting in a contract liability."

As seen in [Exhibit 12](#), AT&T contract assets have been increasing faster than wireless revenue, indicating increasing promotional discounts on mobile devices. During the same period, we see the same trend in Verizon's contract liabilities ([Exhibit 13](#); note that contract liabilities for wireless carriers also include any upfront billing on monthly contracts). T-Mobile also records promotional discounts in contract assets, but has less discount to amortize than its peers.

Exhibit 13: Rising contract liabilities at Verizon is an indicator of promotional activity levels

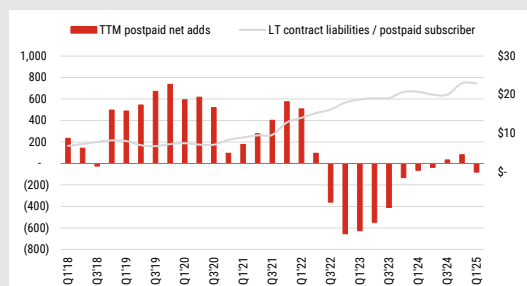


Source: Company filings & Morgan Stanley Research.

Investor Insight: Verizon has said publicly that the amortization of promotional credits should peak as a service revenue growth headwind in 2025, setting up potential service revenue growth acceleration in 2026. A leading indicator for this outcome includes a flattening of contract liability balances on a per postpaid subscriber basis (see [Exhibit 14](#)).

Verizon has been using bill credits as part of its return to subscriber growth. Bill credits per postpaid subscriber have been increasing as net adds turned less negative, and even slightly positive in recent quarters. Because bill credits are amortized through service revenue over the duration of the contract, the use of bill credits to attract and retain subscribers will take time (approximately three years) to amortize through ARPU and income. Verizon has indicated that they expect promo credits to be peaking. This would mean that less amortization would hit revenue in future periods, and cash flow should better track with revenue. The flattening of long-term contract liabilities relative to postpaid subscribers should be a leading indicator for lower bill credit amortization in future periods.

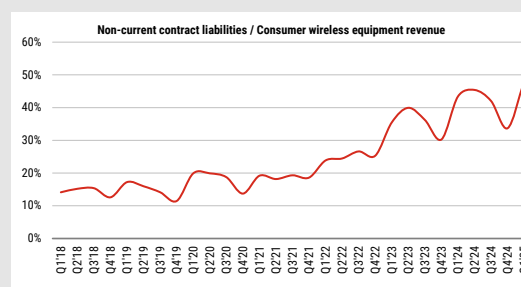
Exhibit 14: A flattening of LT contract liabilities per postpaid subscriber in 2025 would indicate the revenue heading from amortizing promotional discounts will moderate in the future, supporting accelerating service revenue growth acceleration, in line with our published expectations:



Source: Company filings & Morgan Stanley Research.

Another way to look at this trend is by comparing Verizon's non-current contract liabilities (i.e., bill credits) to equipment revenue (note we use only the non-current portion of contract liabilities because the current portion is commingled with standard monthly payments from customers). The amortization of bill credits reduces service revenue over the duration of the installment plan, so we monitor this balance sheet trend as an indicator for future service revenue acceleration.

Exhibit 15: Verizon has been accumulating more bill credits on its balance sheet (approximated using non-current portion of contract liabilities as a % of consumer wireless equipment revenue) that will need to be amortized into revenue over the installment plan



Source: Company filings & Morgan Stanley Research.

Significant Financing Components

Wireless carriers do not charge interest on installment plans, but the contract is considered to have a financing component because the device is delivered well before the company receives payment. Accounting guidance requires companies to recognize the time value of money by calculating the implied interest in the contract.

Verizon has determined that payment plans on devices sold through its direct channel do not have a significant financing component, while T-Mobile has determined that all installment plans contain a significant financing component. Verizon's policy "results in additional revenue recognized upon the sale of wireless devices and no interest income recognized over the device payment term" for direct channel sales (2018 10-K). Verizon's accounting policy will have higher equipment revenue than T-Mobile, and T-Mobile will have a higher margin on equipment sales (due to lower equipment revenue) as compared to Verizon. The companies' filings do not provide enough information to quantify the impact. Note that AT&T does not disclose its accounting policy for the financing components of installment receivables.

Verizon and T-Mobile Financing Component Accounting Policies	
VERIZON 2024 10-K: REVENUE RECOGNITION ACCOUNTING POLICY	T-MOBILE 2024 10-K: SIGNIFICANT ACCOUNTING POLICIES, EQUIPMENT REVENUES
"Based on current facts and circumstances, we determined that the financing component in our existing wireless device payments and fixed-term contracts sold through the direct channel is not significant and therefore is not accounted for separately."	"We recognize as a reduction of the total transaction price the effects of a financing component in contracts via the imputation of interest when customers purchase their devices and accessories on an EIP, including those financing components that are not considered to be significant to the contract."

Equipment Installment Plan Receivables

Wireless carriers offer customers the option to pay for devices over 24 to 36 months through Equipment Installment Plans (EIPs). The payment plans facilitate phone sales and customer retention over the payment term, but create a cash flow gap for the carriers who have to pay for their phone inventory upfront. To address this gap, carriers enter into financing arrangements with third-party financial institutions to securitize these receivables, unlocking the working capital instead of waiting to receive customer payments.

Carriers access both public and private asset-backed securities (ABS) markets using captive finance arms and off-balance-sheet programs. How a carrier chooses to access these markets impacts accounting treatment and affects comparability. **Metrics including leverage, return on invested capital (ROIC), enterprise value, cash collection cycle, and days sales outstanding (DSO) are all influenced by these strategies.**

From a credit perspective, these arrangements introduce debt-like liabilities and structural subordination into the capital structures as cash collected through securitized receivables must be held in separate accounts.

Investor Insight: All three carriers use unique approaches to monetizing EIP receivables. Verizon's approach keeps the receivables on balance sheet and cash collection occurs broadly consistent with the timing of customer collections. As a result, this is a longer cash collection cycle than AT&T and T-Mobile. T-Mobile has typically sold ~25% of its receivables, which are de-recognized from its balance sheet, so it is able to effectively collect cash earlier on those sold receivables. AT&T typically sells as much as 80% of its receivables. Receivable sales impact the timing of cash collection, giving both T-Mobile and AT&T greater control over quarterly FCF generation.

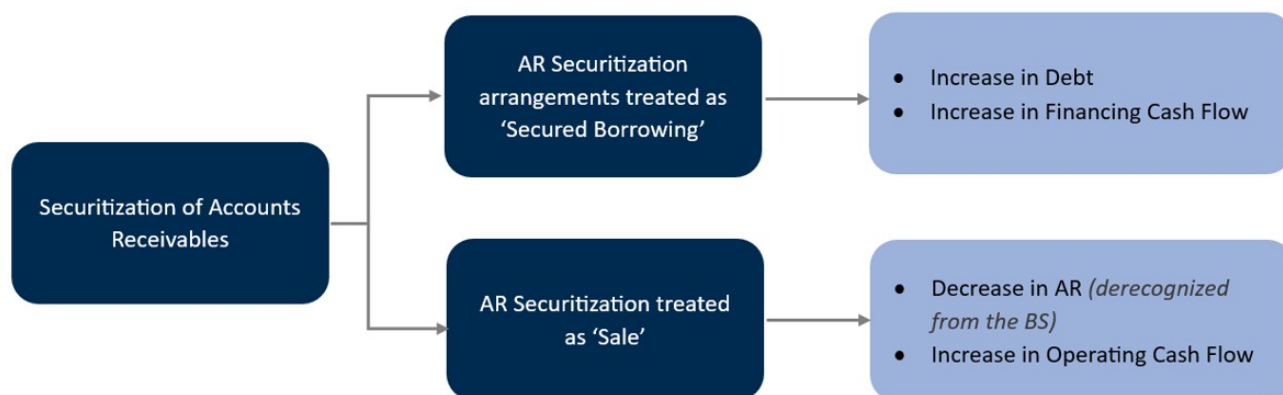
How are securitization arrangements reported on the financial statements?

In order to sell receivables, companies create bankruptcy-remote special-purpose securitization entities. The receivables are transferred to the special purpose entity. The special purpose entity then issues debt securities backed by the receivables. Whether the receivables stay on balance sheet or are derecognized depends on whether the company has fully transferred control.

- *If control has transferred*, the receivables are derecognized, and cash is recognized as an operating inflow.
- *If control is not transferred*, the company keeps the receivables on balance sheet, records a secured loan, and cash from the debt issuance is recorded as a financing inflow. As customers make payments, cash is recorded as an operating inflow.

The diagram below summarizes these securitization structures and associated financial statement impacts:

Exhibit 16: Economically similar transactions receive different accounting treatments:



Source: Morgan Stanley Research

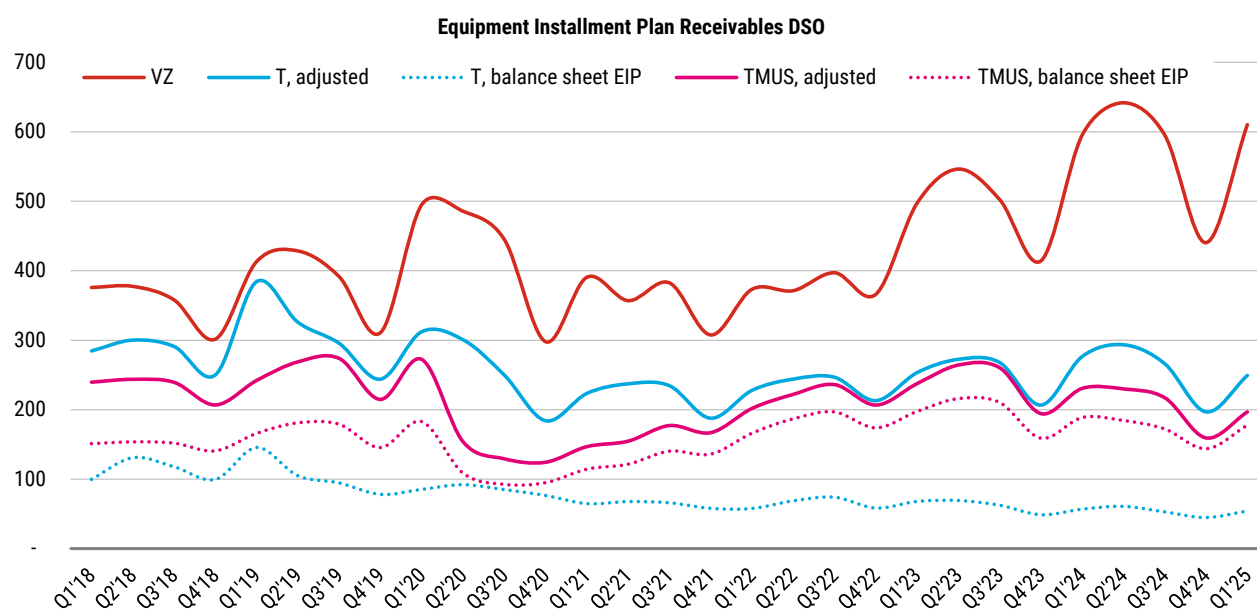
Equipment Installment Receivables Comparability

Verizon's EIP receivables are not comparable to T-Mobile or AT&T for two reasons. First, Verizon records receivables gross of any bill credit discounts or trade in agreements, while T-Mobile and AT&T record receivables net of bill credits (see revenue recognition above). And second, Verizon retains all receivables on balance sheet while T-Mobile and AT&T sell and derecognize a portion of their EIP receivables.

Furthermore, AT&T and T-Mobile EIP days outstanding are not comparable because AT&T sells almost 80% of its EIP receivables compared to T-Mobile's ~30%.

These differences in accounting policy and business practice create challenges in comparing the remaining duration of outstanding contracts across companies or assess collection trends.

Exhibit 17: The chart below highlights the lack of balance sheet comparability among the major operators. AT&T would appear to be the lowest receivables outstanding among the peer group based on receivables on its balance sheet. However, after adjusting for derecognized EIP receivables, AT&T's receivables are higher than T-Mobile. Furthermore, after adjusting for derecognized receivables at both AT&T and T-Mobile, Verizon's installment plan receivables is still substantially higher than AT&T or T-Mobile due to Verizon's accounting policy election to record receivables gross versus its peers electing to record receivables net of bill discounts.



Source: Company filings & Morgan Stanley Research.

EIP Program Impacts Broken Down by Each Wireless Carrier

Verizon

Verizon monetizes its receivables with asset-backed debt. The receivables stay on balance sheet and collections are recorded in operating cash flows. The borrowing is recorded as debt, with proceeds recorded in cash flow from financing.

Verizon's ABS debt balance has grown to more than \$26 billion at 1Q25 end. Increasing installment plan receivables, partially driven by extending installment plans from 24-30 months to 36 months at the beginning of 2022, has allowed the company to secure more asset-backed debt. VZ's asset-backed debt accounts for ~10% of its invested capital. We estimate VZ's EV/EBITDA multiple would be 0.5x lower if we excluded these borrowings, a more consistent comparison to AT&T and TMUS (see [Exhibit 18](#)).

Investor Insight: When Verizon shifted from the private securitization market to on-balance-sheet ABS debt in 2016, management flagged that ABS debt receives favorable captive finance treatment from S&P and Fitch. These agencies treat the securitization as a separate pool of financing, similar to the auto industry. We find many investors exclude ABS debt from leverage metrics, and we would generally agree with this approach when ABS debt is ancillary to an entity's operations. Verizon itself excludes its ABS debt from its net debt and leverage ratios.

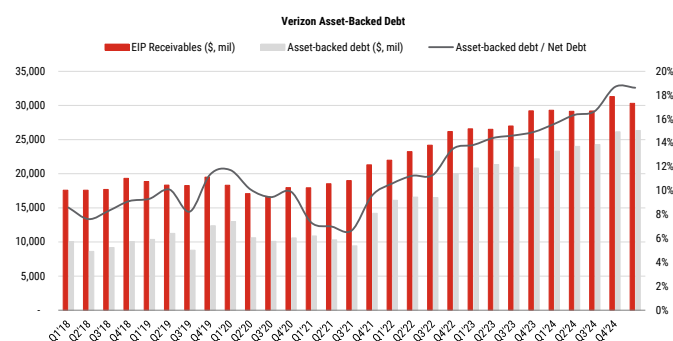
However, we believe the ability to securitize receivables is central to Verizon's operations, so we include ABS debt in EV. This also aligns with Moody's, which adds ABS debt to total debt as an adjustment (see 9/10/24 Credit Opinion). Without access to ABS debt, Verizon would likely have to raise capital at a less favorable rate, which could impact its ability to extend credit to its customers. Furthermore, the exclusion approach ignores the significant risk to the business during a consumer credit downturn. If EIP defaults and delinquencies were to rise, the cost of financing equipment installment plans could increase, causing cascading effects through the P&L from reduced new equipment sales to higher subscriber churn and increased interest expense. We discuss the treatment for AT&T and TMUS EIP receivables, which differs slightly, in [Debt Adjustments by Rating Agencies & Credit Investors](#).

Exhibit 18: If we adjust Verizon EV for ABS debt / EBITDA, VZ's current EV/EBITDA trading multiple would be reduced by approximately half a turn.

	Current Price		
(\$ Millions, except per-share data)	2024	2025E	2026E
Fully Diluted Shares Outstanding	4,214.70	4,223.62	4,226.04
Share Price	\$43.19	\$43.19	\$43.19
Total Market Value	\$182,032.9	\$182,418.0	\$182,522.7
Plus:			
Debt	\$144,014.0	\$140,280.3	\$137,251.5
Other Liabilities	0.0	0.0	0.0
Total Debt	\$144,014.0	\$140,280.3	\$137,251.5
Preferred Stock	\$0.0	\$0.0	\$0.0
Minority Interest	\$1,338.0	\$1,338.0	\$1,338.0
Other Obligations (Pension/OPEB)	\$7,747.6	\$6,913.2	\$5,940.4
Less:			
Cash and Cash Equivalents	\$4,194.0	\$4,405.3	\$6,397.0
Enterprise Value	\$330,938.5	\$326,544.2	\$320,655.7
Less: Asset-Backed Debt	\$26,337.0	\$26,337.0	\$26,337.0
Enterprise Value less Asset-Backed Debt	\$304,601.5	\$300,207.2	\$294,318.7
Total EBITDA	\$48,791.0	\$50,341.9	\$51,886.5
EV / EBITDA	6.8x	6.5x	6.2x
Total EBITDA	\$48,791.0	\$50,341.9	\$51,886.5
EV less Asset-Backed Debt / EBITDA	6.2x	6.0x	5.7x
Forward Total EBITDA	\$50,341.9	\$51,886.5	\$52,987.7
EV / Fwd. EBITDA	6.6x	6.3x	6.1x
Forward Total EBITDA	\$50,341.9	\$51,886.5	\$52,987.7
EV less Asset-Backed Debt / Fwd. EBITDA	6.0x	5.8x	5.6x

Source: Company Filings & Morgan Stanley Research.

Exhibit 19: Verizon has had increasing access to asset-backed debt at favorable borrowing rates as its EIP receivables grow. Cash flows from these EIP receivables are pledged to support the debt.



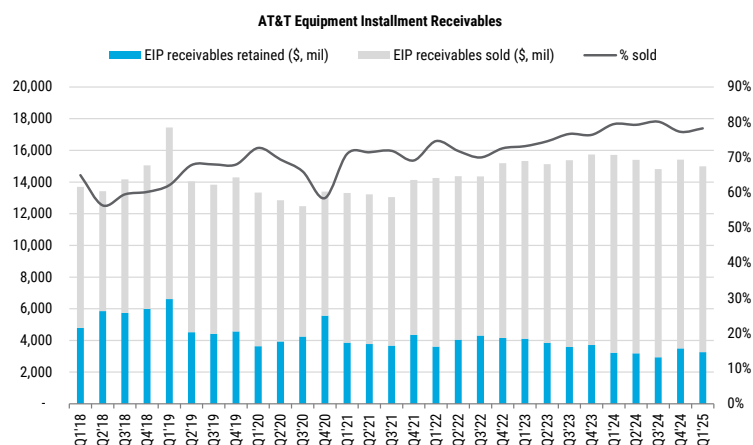
Source: Company filings & Morgan Stanley Research.

AT&T

AT&T uses two securitization programs to monetize its receivables. AT&T's largest program is an off-balance-sheet private securitization program. AT&T reflects cash from these sales in operating cash flow at the time of sale. As seen in [Exhibit 20](#), AT&T sells 70-80% of their equipment installment receivables under this facility.

In 2Q23, AT&T began monetizing additional receivables through a \$1.5 billion revolving credit agreement. Under the revolving receivables agreement, AT&T transfers receivables to a bankruptcy-remote subsidiary and receives cash in the amount of the gross value of the receivables. These receivables stay on-balance-sheet, but the facility allows AT&T to accelerate cash collections. In order to facilitate the transfer, AT&T transfers excess receivables to the subsidiary that are pledged as collateral. In 2024, AT&T expanded the program \$2.7 billion.

Exhibit 20: AT&T sells 70-80% of all EIP receivables



Source: Company Filings & Morgan Stanley Research

Investor Insight: AT&T discloses the change in Equipment Installment Receivables and Related Sales in its cash flow statement. In addition to this disclosure, investors can track cash proceeds received, net of remittances, from sales of EIP receivables in 10-Q and 10-K filings (not provided in earnings releases). The proceeds, net of remittances, from sales of EIP receivables can change substantially from one period to the next, which in turn has a substantial impact on FCF. The amount and timing of EIP receivable sales can be impacted by upgrade cycles, timing of the receivables transfers, and other market conditions. As seen in [Exhibit 21](#), investors can track these proceeds to consider the impact to FCF generation in the period.

Exhibit 21: AT&T cash proceeds received, net of remittances, from sales of EIP receivables

AT&T Sales of Receivables Details (\$mn)	Total since inception			Quarter		
	EIP	Revolv	Sum	EIP	Revolv	Sum
6/30/2016	3,673		3,673			
9/30/2016	3,496		3,496	(177)		(177)
12/31/2016	3,436		3,436	(60)		(60)
3/31/2017	3,740		3,740	304		304
6/30/2017	3,946		3,946	206		206
9/30/2017	4,019		4,019	73		73
12/31/2017	4,337		4,337	318		318
3/31/2018	5,569		5,569	1,232		1,232
6/30/2018	5,723		5,723	154		154
9/30/2018	6,267		6,267	544		544
12/31/2018	6,508		6,508	241		241
3/31/2019	8,387		8,387	1,879		1,879
6/30/2019	7,073	3,725	10,798	(1,314)	3,725	2,411
9/30/2019	6,920	3,456	10,376	(153)	(269)	(422)
12/31/2019	7,211	4,300	11,511	291	844	1,135
3/31/2020	7,156	5,300	12,456	(55)	1,000	945
6/30/2020	6,429	5,300	11,729	(727)	0	(727)
9/30/2020	5,944	5,100	11,044	(485)	(200)	(685)
12/31/2020	5,646	5,300	10,946	(298)	200	(98)
3/31/2021	7,176	5,755	12,931	1,530	455	1,985
6/30/2021	6,735	5,891	12,626	(441)	136	(305)
9/30/2021	6,588	5,562	12,150	(147)	(329)	(476)
12/31/2021	6,644	6,280	12,924	56	718	774
3/31/2022	7,670	6,172	13,842	1,026	(108)	918
6/30/2022*	7,709		7,709	39		39
9/30/2022	8,180		8,180	471		471
12/31/2022	8,519		8,519	339		339
3/31/2023	8,529		8,529	10		10
6/30/2023	8,551		8,551	22		22
9/30/2023	8,918		8,918	367		367
12/31/2023*	9,361	1,500	10,861	443	1,500	1,943
3/31/2024*	9,539	1,800	11,339	178	300	478
6/30/2024	8,947	1,800	10,747	(592)	0	(592)
9/30/2024	8,470	2,770	11,240	(477)	970	493
12/31/2024	8,243	2,770	11,013	(227)	0	(227)
3/31/2025	9,137	2,940	12,077	894	170	1,064

Source: Company filings, Morgan Stanley Research. Note: *The revolving credit agreement was conveyed to WBD upon close of the WarnerMedia/Discovery transaction in April 2022, and then a new program was introduced in 1Q24.

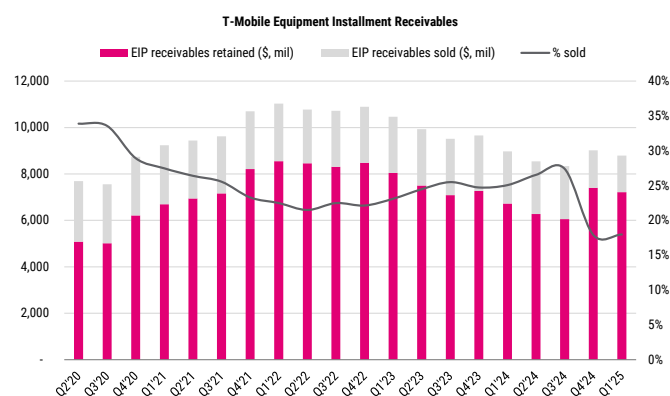
T-Mobile

Similar to AT&T, T-Mobile uses two securitization programs to monetize its receivables. During Q4 2024, T-Mobile amended its off-balance-sheet program. Under the new arrangement, T-Mobile's accounting will converge to that of AT&T with all cash flows from receivables sales in operating cash flows. T-Mobile previously retained a beneficial interest, which meant that significant cash flows from EIP receivables were recorded as investing cash flows and required an additional adjustment to reconcile to FCF.

In addition to selling receivables, T-Mobile issues ABS backed by receivables. Around the time of achieving full investment grade corporate ratings, TMUS began supplementing its securitized receivables program with an ABS program. CFO Peter Osvaldik disclosed at an industry conference in September 2022 that management was looking at other ways to manage the capital structure, including asset-backed securities for equipment receivables.

TMUS did its inaugural ABS deal in October 2022, in connection with which it formed a related wholly owned bankruptcy remote entity and a trust. As of 1Q25 end, the fair value of the ABS Notes to third parties stood at \$1.9bn across four tranches, which were rated **Aaa** and **AAA** by **Moody's** and **Fitch** respectively. Net proceeds from ABS issuance are captured in Proceeds from issuance of long-term debt under Financing activities on the cash flow statement, and the ABS notes are included in the balance sheet debt balances.

Exhibit 22: T-Mobile EIP receivables sales have been declining over time, which could be due to higher bill credits netted against the receivables and/or slowing upgrade cycle, and the company is retaining more of the outstanding receivables on balance sheet



Source: Company Filings & Morgan Stanley Research.

Investor Insight: Historically, T-Mobile sold its installment receivables to a third party but retained a beneficial interest. The proceeds at the time of sale were recorded as operating cash inflows. As collections from the beneficial interest were received, the collections were recorded as investing cash inflows. T-Mobile's free cash flow measure adjusted for the impact of the beneficial interest accounting in investing cash flows ([Exhibit 23](#)). Although this adjustment will not be required going forward, investors should be aware that they need to include these cash flows in historical calculations.

Exhibit 23: T-Mobile's historical Free Cash Flow calculation includes an adjustment for collections on their retained interest in EIP receivables that were sold

Adjusted Free Cash Flow is calculated as follows:

(in millions, except percentages)	Quarter								Year Ended December 31,	
	Q1 2023	Q2 2023	Q3 2023	Q4 2023	Q1 2024	Q2 2024	Q3 2024	Q4 2024	2023	2024
Net cash provided by operating activities	\$ 4,051	\$ 4,355	\$ 5,294	\$ 4,859	\$ 5,084	\$ 5,521	\$ 6,139	\$ 5,549	\$ 18,559	\$ 22,293
Cash purchases of property and equipment, including capitalized interest	(3,001)	(2,789)	(2,424)	(1,587)	(2,627)	(2,040)	(1,961)	(2,212)	(9,801)	(8,840)
Proceeds from sales of tower sites	6	2	2	2	—	—	—	—	12	—
Proceeds related to beneficial interests in securitization transactions	1,345	1,309	1,131	1,031	890	958	984	747	4,816	3,579
Adjusted Free Cash Flow	\$ 2,401	\$ 2,877	\$ 4,003	\$ 4,305	\$ 3,347	\$ 4,439	\$ 5,162	\$ 4,084	\$ 13,586	\$ 17,032

Source: Company filings & Morgan Stanley Research.

Charter Communications

Charter recently started testing the handset receivables waters as well. Charter Communications has provided its Spectrum Mobile service as a mobile virtual network operator (MVNO) using Verizon's network for years now, building to more than 10mm total mobile lines at 1Q25 end.

In June 2024, a bankruptcy remote special purpose vehicle of the company entered into a senior secured revolving credit facility to

finance the purchase of equipment installment plan receivables with a number of financial institutions. As of YE24, the carrying value of the EIP Financing Facility was \$1.1bn and is included as a separate line item in the company's consolidated balance sheets. From the credit perspective, as of YE24, we adjust the reported total debt by \$1.8bn to include that balance along with the short-term borrowings under the company's supply chain finance programs.

Supplier Financing Programs

Vendor and supplier financing programs can cause fluctuations in reported cash flows and shift cash flows associated with certain payables from operating to financing. AT&T and Verizon to a lesser extent use these programs to extend payables and manage working capital. In our FCF forecasts, we include vendor financing payments, in line with common practice among the major US operators. We also monitor the usage of these programs closely as they can have a significant impact on reported cash flow.

What Are Supplier Financing Programs? Supplier financing, also known as reverse factoring or supply chain financing, is a financing arrangement between customers, suppliers and banks that improves working capital.

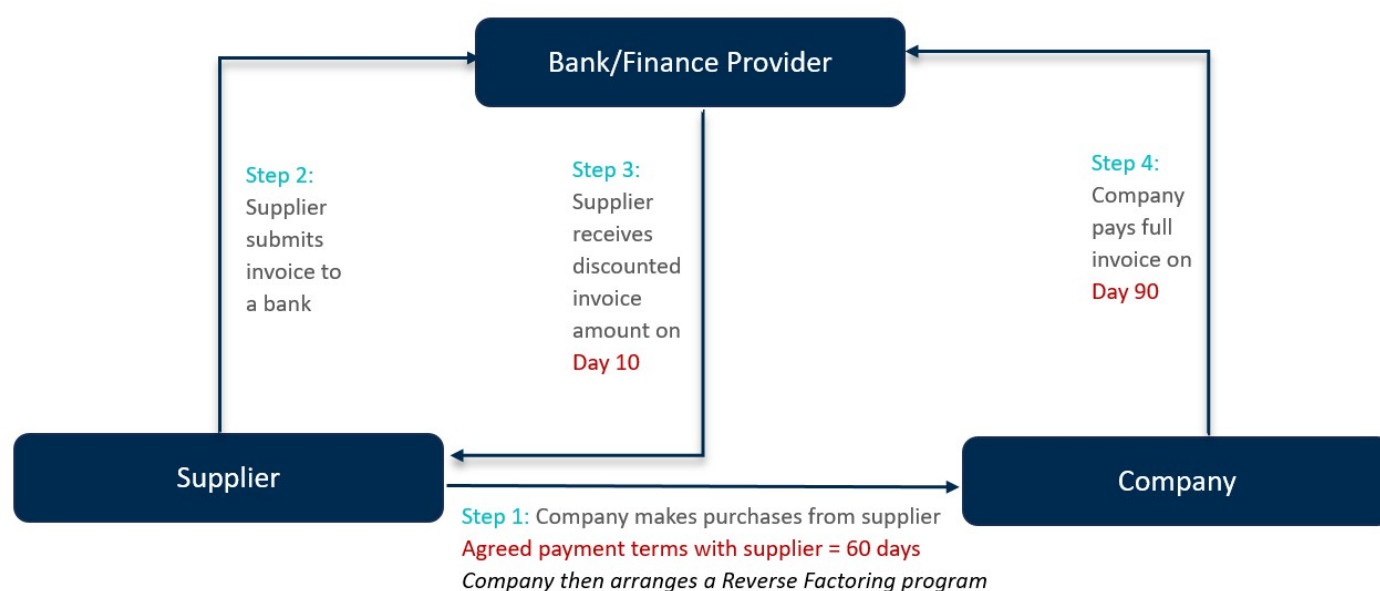
In a traditional transaction, a company makes purchases from a supplier and records Accounts Payable on the balance sheet. The company then pays the supplier according to the agreed upon terms — which are typically determined based on buyer-supplier relationship and/or industry standards. Accounts Payable grows as new pur-

chases are made and decreases as invoice payments are made to suppliers.

With supplier financing programs, the company enters into a program with bank that enables its suppliers to gain immediate funding while the company receives extended payment terms with the bank. Consider the following illustrative example:

- **Step 1:** The company makes purchases from a supplier. The supplier sends the inventory along with an invoice with the agreed upon payment term at the time of 60 days.
- **Step 2:** Because the company has a supplier financing program with a bank, the supplier presents the invoices to the bank, which pays the supplier on day 10 at a small discount.
- **Step 3:** Depending on the arrangement with the bank, the customer may also be able to extend the payment terms beyond the original 60 day terms, paying the bank on day 90. The extended payment terms create a source of short term funding the company, improving its cash flow.

Exhibit 24: Below is an illustrative example of a supplier financing arrangement



Source: Morgan Stanley Research

Beginning in 2023, new accounting rules required that companies provide more disclosures about supplier financing programs. As a result of these new rules, Verizon began providing disclosures about its vendor financing program, and AT&T enhanced its existing disclosures. T-Mobile has disclosed supplier financing programs in the past, but at this time is not providing any disclosure, so it is likely not using such financing arrangements.

Verizon extends its payment terms from 60 days to 90 days under its supplier finance arrangement with a third party financial institution. At the end of 2024, (earliest disclosure provided), VZ had ~\$800 million of obligations outstanding under the program. Supplier usage of the program has fluctuated between \$600 million to \$1 billion over the last two years.

AT&T aims to pay vendors on 90 day payment terms or longer. In order to extend these payment terms, AT&T uses (1) a supplier financing program with a third party institution, (2) a direct supplier financing program, and (3) a vendor financing program.

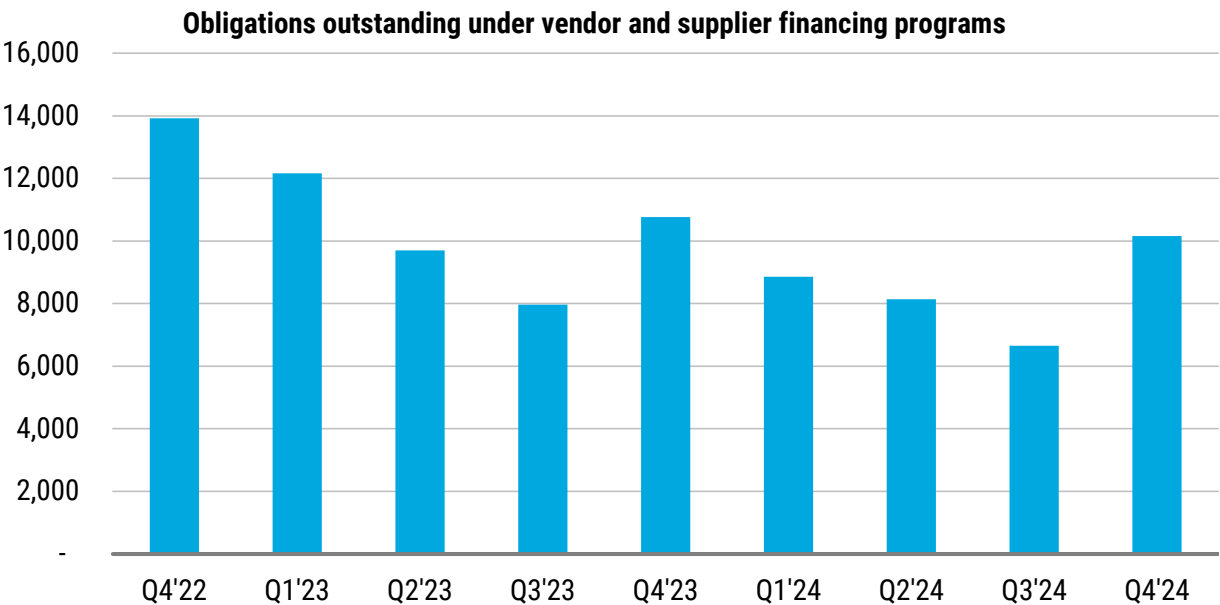
- *Supplier financing program:* AT&T provides suppliers with access to bank facilities that allow suppliers to collect payments earlier, at the supplier's cost. The supplier financing

program does not result in changes to normal payment terms or cash from operations. Usage of the bank facility by suppliers has declined from \$2.9 billion in 2022 to \$2.5 billion at YE 2024.

- *Direct supplier financing program:* AT&T can extend the agreed upon payment terms by up to 90 days, at an additional cost to AT&T. This program increases operating cash flow in periods where AT&T extends more invoices and decreases operating cash flow in periods where AT&T uses less. AT&T's usage of the program has been increasing over time, from \$5.5 billion in 2022 to \$6.3 billion at YE 2024.
- *Vendor financing program:* This program is used to negotiate payment terms of 120 days or more for purchases of capital assets. Payments under this program are recorded as financing outflows in the statement of cash flows. Usage of the program peaked in 2022 at \$5.6 billion and has declined to \$1.4 billion at YE 2024.

AT&T has stated its intent to reduce usage of the direct supplier financing program and vendor financing programs over time, which they manage together. [Exhibit 25](#) below shows AT&T's aggregate obligations outstanding under all three of these programs over time, which have been used to manage as much as \$14 billion of payables.

Exhibit 25: AT&T Supplier and Vendor Financing Programs have been used to manage as much as \$14 billion of payables

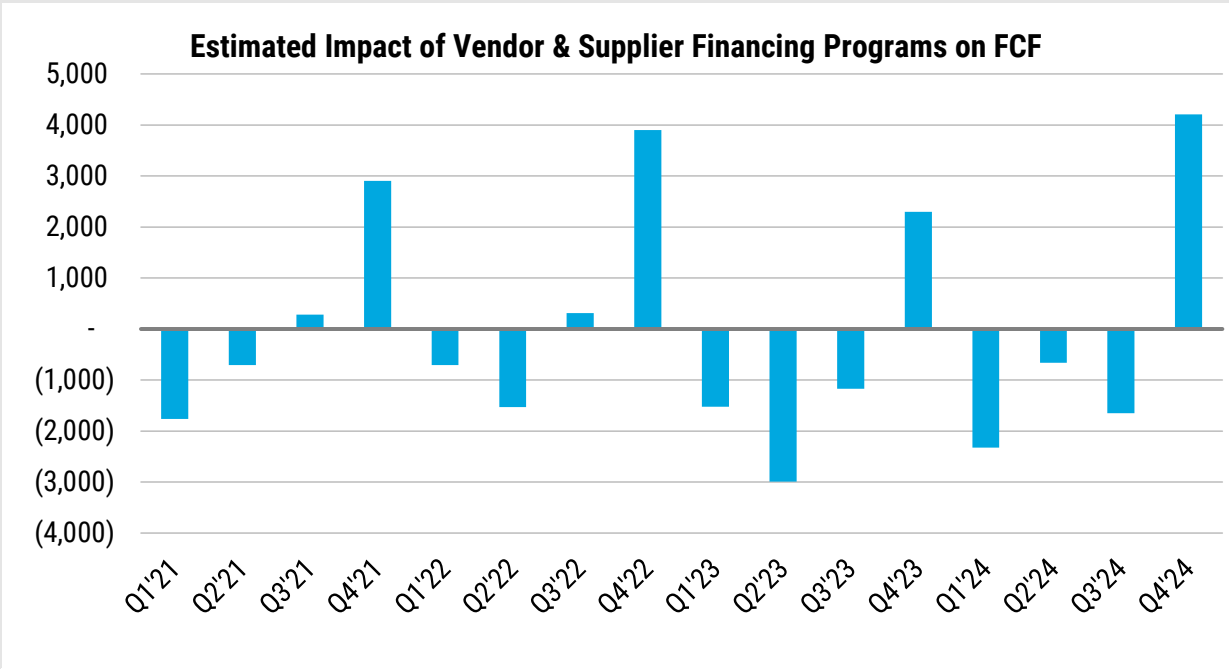


Source: Company filings & Morgan Stanley Research.

Investor Insight: Investors should be aware of the usage of supplier financing facilities because they can have a large impact on free cash flow. [Exhibit 26](#) below shows the quarterly FCF impact of vendor and supplier financing programs. AT&T had been increasing usage of the facilities in 2021 and 2022, resulting in a positive annual FCF impact in those years. In 2023, AT&T significantly reduced usage, reducing FCF by ~\$3.4 billion that year, and further reduced usage throughout 2024, reducing FCF that year by ~\$0.4 billion.

While it is challenging to forecast AT&T's future usage of supplier financing programs, investors should be aware that these cash flows are irregular when considering the sustainability of reported FCF in any given period.

Exhibit 26: AT&T's vendor and supplier financing programs have a large impact on FCF from one period to the next. When usage is increasing, reported cash flow is higher. When usage decreases, cash flow declines.



Source: Company filings & Morgan Stanley Research.

Spectrum Licenses - The Accounting & Tax Gift to Wireless Carriers

The accounting and tax treatment for spectrum licenses is highly beneficial to wireless operators. Spectrum assets are expensed for tax purposes but not amortized under accounting rules, and investments are reported separately from other capital expenditures. As a result, spectrum licenses, a core operating asset, are not directly factored into either EPS or FCF.

Spectrum purchases are largely driven by the FCC's auction schedule, making cash flow forecasting difficult. However, ongoing investment in spectrum is essential for deploying wireless services, we incorporate spectrum cost in our 10-year DCF analysis.

Spectrum Accounting Treatment

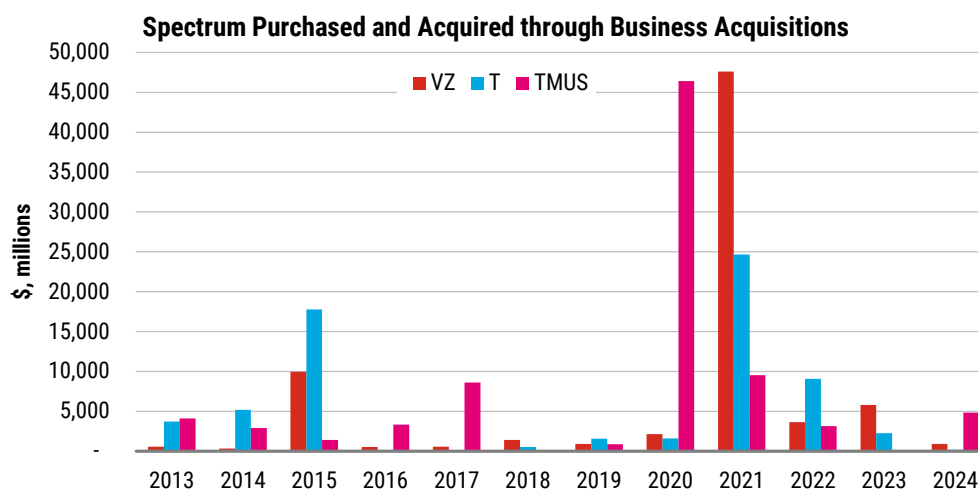
Wireless communication services are provided on exclusive spectrum licenses purchased from the federal government. The Federal

Communications Commission (FCC) auctions spectrum licenses that grant the winner exclusive rights to specific radio frequencies in a geographic area, generally renewable into perpetuity. Because spectrum auctions occur at the discretion of the FCC, companies do not have much, if any, control over the timing of these investments. For example, the FCC auctioned over \$80 billion of 5G spectrum in 2021. That year, Verizon won over \$47 billion of these licenses, funded mainly with new debt, after not making any substantial investment in new spectrum for five years. T-Mobile acquired spectrum valued at nearly \$45 billion when it acquired Sprint in 2020.

Because spectrum licenses have an unlimited useful life, they are treated as indefinite-lived intangible assets for accounting purposes. As such, they are not amortized and remain on the balance sheet at their historical cost unless there is an impairment.

Investor Insight: Spectrum purchases are quite lumpy over time as compared to other types of capital expenditures ([Exhibit 27](#)). However, wireless carriers will have to continue to make spectrum purchases in the long-run in order to provide wireless services, deploy new technologies, and meet data usage demands. Due to the lumpy nature of the capital deployment, some investors exclude spectrum purchases from free cash flow measures. Given that these investments must be made in perpetuity in order to remain competitive, investors should consider these ongoing investments when modeling future cash flows and in determining sustainable free cash flow margins.

Exhibit 27: T-Mobile acquired significant spectrum assets with Sprint in 2020, while Verizon and AT&T made significant purchases in the 2021 FCC auction



Source: TMUS 2020 spectrum acquired in Sprint acquisition.
Company filings & Morgan Stanley Research.

Spectrum License Tax Treatment

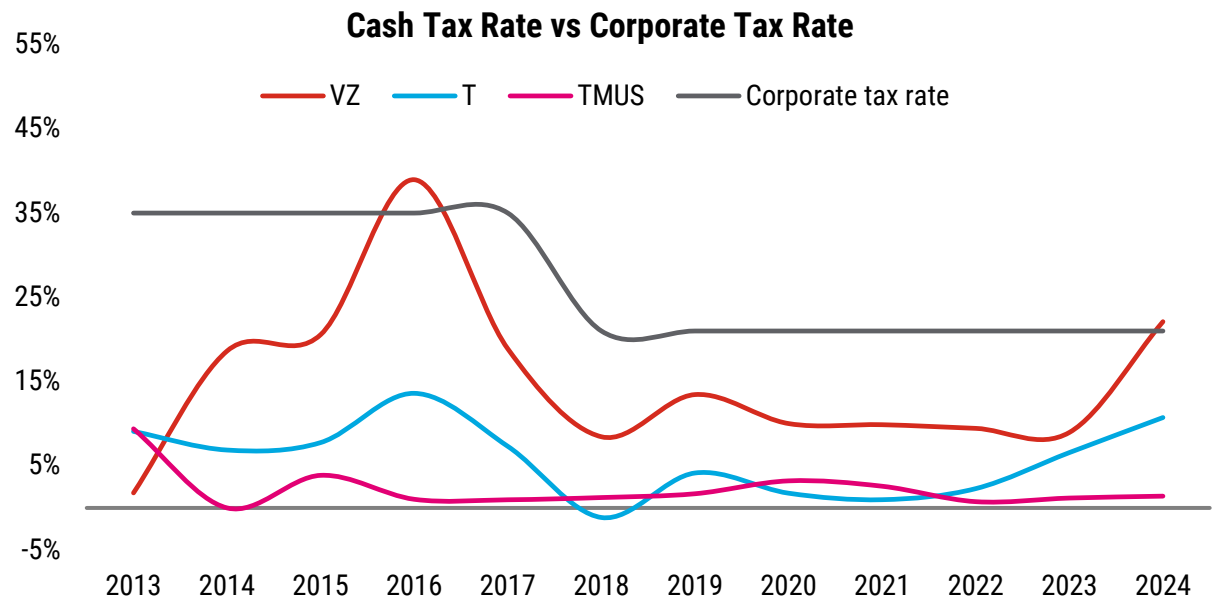
For income tax purposes, spectrum licenses are amortized on a straight-line over a 15 year period according to the amortization rules for intangibles, reducing cash tax expense throughout the period.

Because spectrum is amortized for tax purposes and not under GAAP, a deferred tax liability accrues on the balance sheet as the tax basis of the spectrum declines over time. This deferred tax liability will remain on the balance sheet unless the spectrum is sold to another entity. In the event that spectrum is sold, the company would be

required to pay taxes on the difference between the selling price and the tax basis (if the spectrum was fully amortized for tax purposes, the tax basis would be zero and the entire value of the transaction would be taxable upon sale).

The tax shield from spectrum licenses can be quite large. As a result, [Exhibit 28](#) shows the cash tax rate paid by wireless carriers, which has been substantially lower than the statutory corporate tax rate.

Exhibit 28: The major wireless carriers have paid a significantly lower cash tax rate than the corporate tax rate due to spectrum amortization and tax incentives such as bonus depreciation (see below)



Source: Company filings and Morgan Stanley Research. Earnings before income taxes adjusted for company non-GAAP adjustments.

Spectrum Capitalized Interest

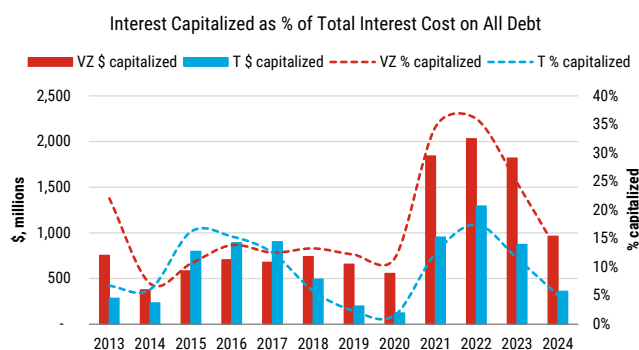
Interest expense on debt incurred to acquire spectrum licenses is capitalized and added to the intangible asset cost basis on the balance sheet up until the spectrum is deployed. Because the intangible asset has an indefinite life under GAAP, this capitalized interest will not amortize into the income statement.

After winning substantial spectrum licenses at the 2021 auction, both AT&T and Verizon were capitalizing a large portion of their interest expense (Exhibit 29). Of the total amount of interest capitalized by these companies in the last three years, over 80% has been capitalized into wireless licenses on the balance sheet. AT&T capitalized \$2.8 billion to the spectrum license intangible since 2021, and Verizon capitalized \$5.3 billion during the same period.

Once spectrum is placed into service, the interest expense can no longer be capitalized. As seen in Exhibit 30, capitalized interest declined significantly in 2024 as the purchased spectrum has been placed into service.

We include interest capitalized in spectrum in our free cash flow calculation to capture the full cost of debt.

Exhibit 29: Verizon and AT&T capitalized billions of interest expense into spectrum assets in recent years, which we include in free cash flow measures



Source: Company filings & Morgan Stanley Research.

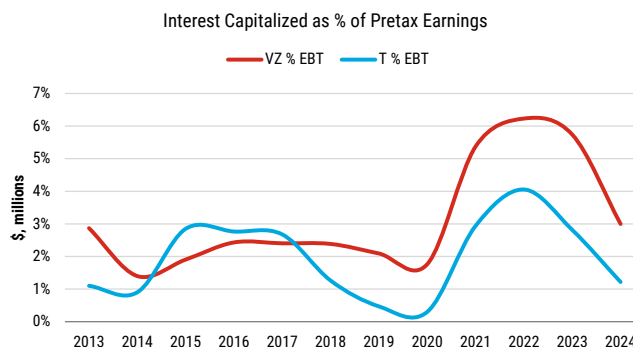
Spectrum accounting treatment and relative valuation

In our view, the accounting treatment and investment norms around mobile carriers overstates the earnings and FCF generation of these businesses. This is because the acquisition of spectrum licenses, a core operating asset, is not factored into either EPS or FCF.

If we compare wireless carriers to cable or other fixed-line operators, there is a clear inconsistency in how the accounting rules treat the core operating assets of each group.

When a cable operator needs to add capacity to its network, it deploys capital, and that capital is capitalized and depreciated against earnings. That capital outflow is reported as capital spending, reducing FCF generation. In other words, the EPS and FCF estimates for Comcast, Charter, and Altice all reflect historical and future expected investments in their plant.

Exhibit 30: Capitalized interest has historically been 1-5% of pretax earnings for VZ and T



Source: Company filings & Morgan Stanley Research. Pretax income adjusted for company Non-GAAP adjustments.

When a wireless carrier acquires spectrum licenses, that asset is capitalized as an intangible asset with an infinite life. This infinite life treatment comes despite the fact that technically speaking, these are not "perpetual licenses" and remain under the authority of the FCC. Of note, the carriers do get to depreciate their spectrum acquisitions against their cash taxes, so it's "all gain, no pain" from an accounting point of view.

As a result of this accounting treatment, there is no depreciation of spectrum against earnings for mobile carriers like AT&T, Verizon, and T-Mobile. In addition, investors do not burden FCF expectations explicitly for future spectrum needs. The timing and magnitude of new spectrum supply is unknown, making forecasting challenging. The only analysis that picks up the asset intensity of the business is a ROIC analysis, which we discuss below.

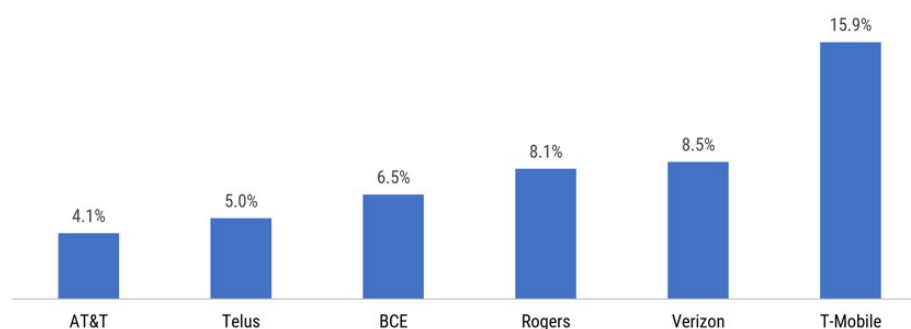
The implication of all this is that comparing the P/E of CMCSA and CHTR to that of TMUS, VZ, and T are not apples-to-apples. All else equal, the wireless carriers should trade lower as the cost associated with existing and future spectrum needs are not reflected in their earnings.

Our approach to forecasting future spectrum needs:

Implicitly, the trading multiples for the wireless carriers appear to reflect some expectation of future spectrum capital spending, although it is impossible to calculate an exact expectation, in our view. Our approach across our coverage group is to 1) look at the historical growth in spectrum assets on the balance sheet ([Exhibit 31](#) and [Exhibit 32](#)), and 2) to assume spectrum capital spending in the future that reflects some future growth in spectrum asset.

Exhibit 31: All the major carriers have increased their investment in spectrum over time, including inorganically through acquisitions like T-Mobile's Sprint acquisition

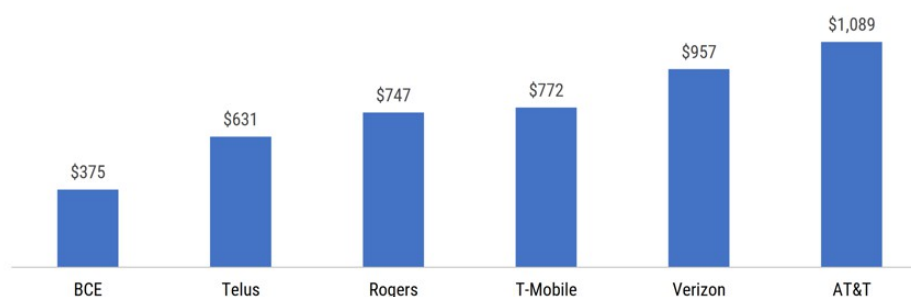
Spectrum Asset CAGR Across US Wireless and Canadian Wireless (2017-2024E)



Source: Morgan Stanley Research estimates . Note: T-Mobile includes the acquisition of Sprint (closed in 2020). All currencies for Canadian players have been converted to USD at 1.40 CAD/USD rate

Exhibit 32: We see a range of invested capital in spectrum relative to current wireless customer bases across the industry

Spectrum Asset / EOP Customer Across US Wireless and Canadian Wireless (2024E)



Source: Morgan Stanley Research estimates . Note: All currencies for Canadian players have been converted to USD at 1.40 CAD/USD rate

For consistency and simplicity sake, we assume a 5% increase in spectrum assets over time in the future across the coverage group – beyond spectrum acquisitions already announced. While these cash outflows are in our long-term forecast and are not impacting 2025 or 2026 results, typically, they do impact future financial capacity for other investments including share buybacks. They also put downward pressure on future ROIC levels. Finally, we also feed this capital spending into our 10-year DCF analyses, which impacts our DCF-driven valuation work.

Finance Leases

Wireless carriers are capital intensive businesses, spending substantial amounts on infrastructure in the form of both capital expenditures and leased assets. In order to account for the full capital intensity of these businesses and to compare across companies that use varying levels of leases, we include lease payments in measures of free cash flow and lease obligations in adjusted debt balances.

Lease Accounting Overview

Lease accounting rules vary between US GAAP and IFRS, resulting in different financial statement impacts for economically similar transactions. Under both standards, a lease right of use asset and corresponding liability is recorded on the balance sheet for all leases (with some exceptions for short duration and low value leases). However, there are major differences on the income statement and cash flow statement that inhibit comparability.

US GAAP accounting requires leases to be classified as either operating leases or finance leases. A lease is classified as a finance lease if the lease term makes up a majority of the economic life of the asset or if the present value of lease payments is substantially all of the asset's fair value. The finance lease model is meant to treat the transaction as though the lessee has effectively purchased the asset and financed that purchase through the lease agreement.

Under the operating lease model, rent expense is recorded in the P&L and lease payments are classified as operating cash flows. In contrast, finance lease expenses are recorded as amortization expense (related to the right of use asset) and interest expense (related to the lease liability) in the P&L. Interest payments are classified as operating cash flows (with the option to present as financing cash flows under IFRS), and principal payments are classified as financing cash flows.

All leases are finance leases under IFRS, and lease-related expenses are recorded as amortization and interest expense. As a result, standard cash flow metrics for lease-intensive IFRS reporters may be overstated relative to their US GAAP counterparts.

We include finance lease principal payments in free cash flow to account for the cost of leased assets and to make cash flow apples-to-apples across companies regardless of lease versus buy decisions and IFRS/GAAP accounting differences.

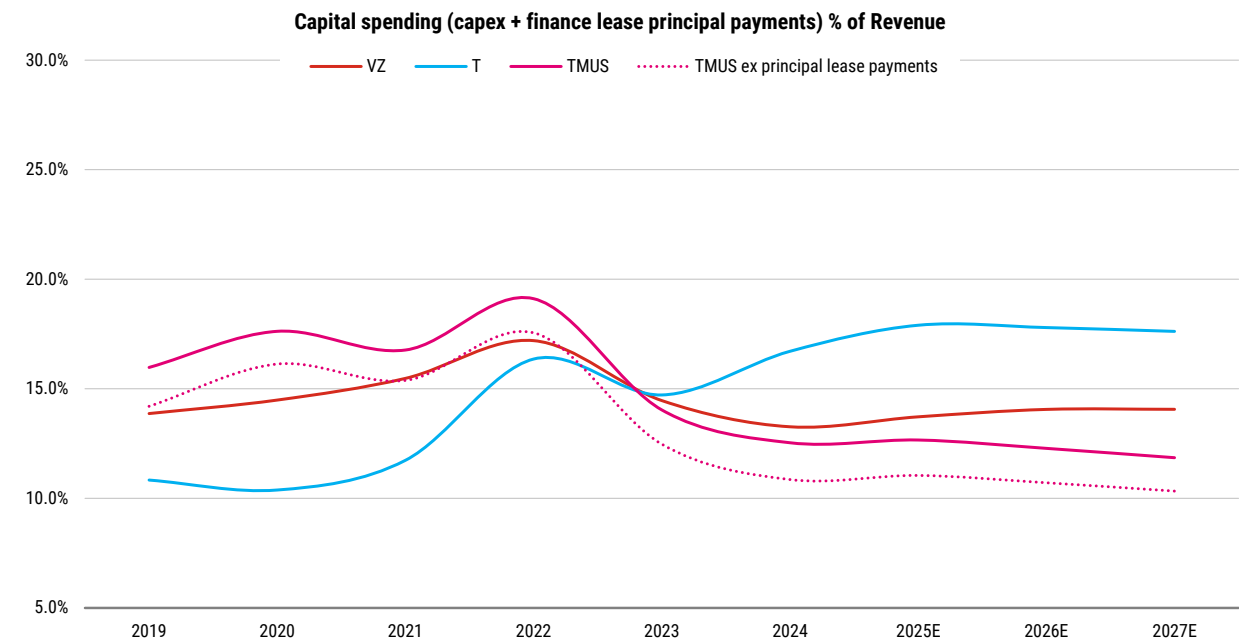
Finance Lease Impact on Measures of FCF and Capital Spending

Companies can finance capital expenditures in a number of ways - using debt, directly negotiating extended payment terms with the vendor (see AT&T's vendor financing program above), or by entering into a finance lease, among others. Under any of these options, the asset is acquired before the cash outlay. Under the first two scenarios, the capital expenditure is captured within free cash flow measures (note that AT&T includes vendor payments in its free cash flow). However, when a company enters into a finance lease, only the interest expense on the finance lease is captured in free cash flow, but the rest of the cost of the asset is not. Principal payments on the finance lease are captured in the financing section. If free cash flow is not adjusted to capture finance leases, free cash flow will not reflect the full capital intensity of the business.

Verizon, AT&T and T-Mobile all use finance leases to acquire capital equipment, but T-Mobile uses finance leases more than its two peers. If investors were to ignore finance lease principal payments in measures of capital intensity, T-Mobile capital intensity is understated by ~10% relative to peers. [Exhibit 33](#) below shows how TMUS capital spending, defined as capex plus principal payments on finance leases, has historically been higher than peers. Without accounting for these principal payments, capital spending would appear in-line with peers, understating the capital intensity of the business, and overstating free cash flow generation.

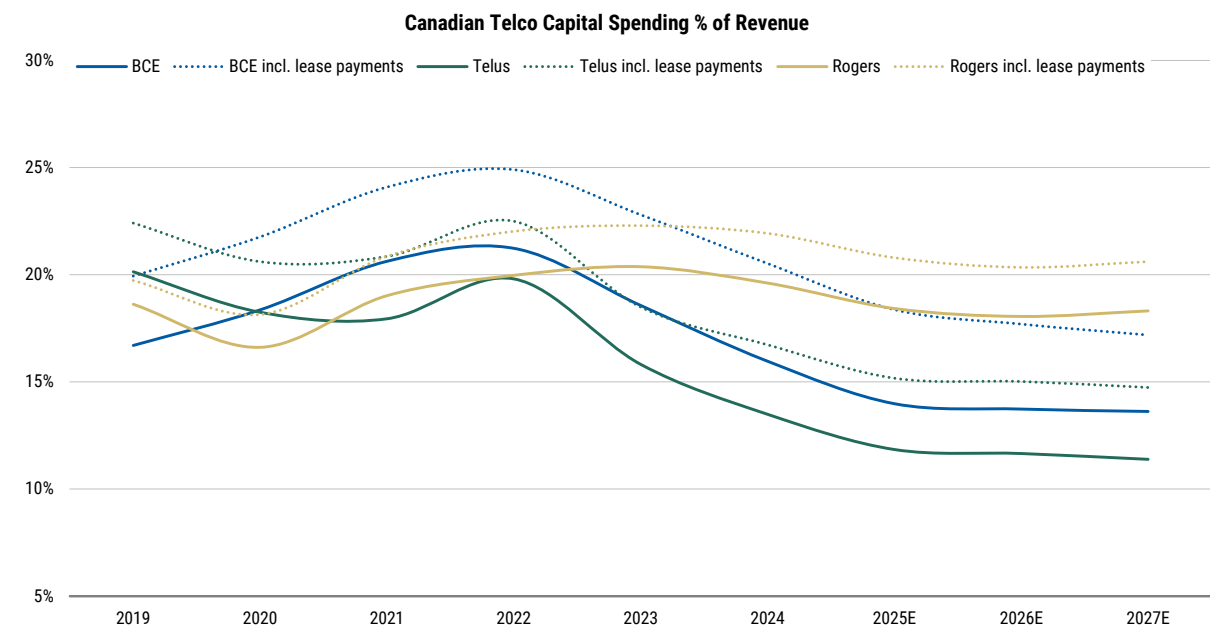
We also call attention to the Canadian telcos, BCE Inc. [BCE], Telus Corp. [T.TO], and Rogers Communications, Inc. [RCIb] which all report under IFRS. Under IFRS, all leases follow the finance lease model, so lease payments are included in finance lease payments within the financing section of the cash flow statement. [Exhibit 34](#) below shows total capital spend at Rogers and BCE is higher than US the peers. We include these payments in our FCF measures to capture the full cost of capex.

Exhibit 33: TMUS spends ~150bps of revenue on finance lease principal payments



Source: Company filings & Morgan Stanley Research.

Exhibit 34: Canadian Telecom capex is significantly understated if principal on lease payments is excluded from free cash flow



Source: Company Filings & Morgan Stanley Research.

For a deeper dive into the Canadian Telecom industry, see our 2025 Year-Ahead Outlook: When Will Winter End for Canadian Telecoms? [here](#).

Tower Obligations and Failed Sale Leaseback Transactions

The wireless carriers monetize their tower assets through sale leaseback transactions with tower companies. Under these arrangements, the wireless carrier receives cash for the tower asset and enters into a long-term lease arrangement with the tower company to continue to use the towers in its operations. Because the wireless carrier has continuing involvement with the tower assets and often retains an option to repurchase the towers in the future, the sale may not qualify as a sale according to GAAP.

Continuing involvement can result in what is called a "failed sale leaseback," where the tower asset remains on the wireless carrier's balance sheet and is depreciated as though the carrier remains the legal owner. The lease is accounted for as a financing transaction,

with the proceeds from the "sale" recorded as a liability that is paid down over time through the rent payments.

Failed sale leaseback accounting requires that the finance obligation cannot exceed the carrying value of the tower at the end of the lease term, which can result in an artificially high interest rate. For example, T-Mobile uses an 11.6% interest rate for some of its tower obligations, much higher than the 4% effective interest rate on its long-term debt ([Exhibit 35](#)). In addition to artificially high interest expense, the high interest rate also means that the tower obligation liability on the balance sheet is paid down more slowly because more of the lease payment is applied to interest expense than to principal.

Exhibit 35: TMUS Tower Obligations 2024 10-K Footnote

Leaseback Arrangement

On January 3, 2022, we entered into an agreement (the "Crown Agreement") with CCI. The Crown Agreement extends the current term of the leasebacks by up to 12 years and modifies the leaseback payments for both the Existing CCI Tower Lease Arrangements and the Acquired CCI Tower Lease Arrangements. As a result of the Crown Agreement, there was an increase in our financing obligation as of the effective date of the Crown Agreement of approximately \$1.2 billion, with a corresponding decrease to Other long-term liabilities associated with unfavorable contract terms. The modification resulted in a revised interest rate under the effective interest method for the tower obligations: 11.6% for the Existing CCI Tower Lease Arrangements and 5.3% for the Acquired CCI Tower Lease Arrangements. There were no changes made to either of our master prepaid leases with CCI.

The following table summarizes the balances associated with both of the tower arrangements on our Consolidated Balance Sheets:

(in millions)	December 31, 2024	December 31, 2023
Property and equipment, net	\$ 2,069	\$ 2,220
Tower obligations	3,664	3,777
Other long-term liabilities	554	554

Future minimum payments related to the tower obligations are approximately \$380 million for the 12-month period ending December 31, 2025, \$788 million in total for both of the 12-month periods ending December 31, 2026 and 2027, \$835 million in total for both of the 12-month periods ending December 31, 2028 and 2029, and \$3.7 billion in total thereafter.

Source: Company filings & Morgan Stanley Research.

Pensions and Other Post-Employment Benefits

The accounting treatment for pensions and other postretirement benefits (OPEB) can result in income even as no cash is received. For valuation, we adjust EPS to exclude forecasted pension earnings. These earnings are non-cash, non-operating, and non-recurring, and as such, we do not put a multiple on them. Instead, our forecasts focus on the actual cash impact from contributions to pension funds and payments for other postretirement benefits, which directly affect operating cash flows. We include the tax-adjusted unfunded portion of pension and other postretirement benefits in our EV calculations.

Background

While defined benefit pensions and other post-employment benefits (OPEB plans) are not as prevalent today due to the challenges of managing such long-duration plans and the rising popularity of alternatives, both AT&T and Verizon employ a large number of unionized employees and continue to provide these benefits. T-Mobile employees are not unionized, and the company does not provide pensions or other postretirement benefits. Investors should care about pensions and post-employment benefits due to the risk associated with underfunding and the potential impact these promised retirement benefits can have on financial results.

Pensions: Pension obligations are contractual and legally enforceable, and ERISA mandates minimum funding requirements. The IRS provides tax benefits for pre-funding pension obligations, meaning that companies receive tax deductions when they contribute funds to the pension trust.

Other Post-Employment Benefits: An OPEB plan pays for retiree healthcare, life insurance, and other post-retirement costs. Since the plan is a defined benefit plan, the risk is borne by the company. Unlike pension plans, OPEB obligations are typically non-contractual; however, investors would likely want to include these liabilities in debt if they expect the company to follow through on its commitment to retirees. Most OPEB plans are "unfunded," meaning that they are not set up in trusts and instead paid out of company cash. ERISA does not regulate OPEB plans, and the IRS generally does not allow deductions for pre-funding the obligations, with certain exceptions.

Accounting for Retirement Benefits

The basic principle behind accounting for retirement benefits is that the employer's obligation to pay post-retirement benefits is a form of compensation, and that obligation should be recognized over the employee's term of service. It is important to note that the cost of the retirement benefits includes the present value of future benefit payments, interest on the retirement benefit obligation less return on

invested plan assets, and amortization of deferred gains and losses. Deferred gains and losses arise from the changes to projections used to value the defined benefit plan obligations, such as changing the discount rate.

On the Balance Sheet The Projected Benefit Obligations and Plan Assets are reported net on the balance sheet. Since benefit obligations are tax-deductible, there is also a deferred tax asset related to the plans.

On the Income Statement The following pension items appear on the income statement:

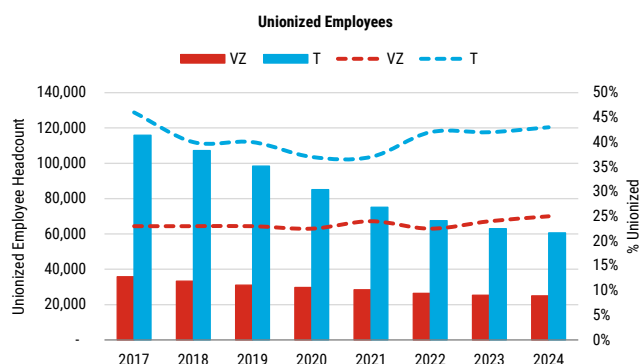
- **Service Cost.** The present value of new benefits earned during the period, represents the economic operating cost of the ongoing plan.
- **Interest Cost.** The corporate AA bond yield multiplied by the projected benefit obligation, represents the financing cost of the gross liability.
- **Expected Return on Assets.** Management's assumption for the long-term return on the portfolio of plan assets. It considers the asset mix, but there is a wide variance of expected return on plan assets even among plans with similar asset allocation.
- **Amortization of Prior Service Costs.** When a plan is amended to change benefits, the projected benefit obligation is re-measured to reflect the new benefit formula. These differences are smoothed into pension expense (amortized) over the remaining service life of employees.
- **Amortization of Actuarial Gains and Losses.** When the actual results of the plan differ from the actuarial estimates, these differences can either be smoothed into pension expense (amortized) over the remaining service life of employees, or can be recognized in income in the period in which the occur.
- **Other.** There are several other items that companies may report in pension/OPEB expense, including settlements and curtailments.

On the Cash Flow Statement Pension and OPEB expense is entirely non-cash. For pensions, benefit payments are paid out of a trust, so they do not appear on the cash flow statement. Cash contributions from the company to the trust may be the minimum required by regulators (ERISA), or may be discretionary, and are included in cash flow from operations. OPEB are typically unfunded, so contributions to the plan are typically immaterial and benefits are paid mainly from operating cash flow.

Retirement Benefits Provided by Verizon and AT&T

AT&T and Verizon each have large unionized workforces and offer defined pension and other postretirement benefits to these employees. Both companies have been scaling down their workforce over time, but as seen in [Exhibit 36](#), both companies still have tens of thousands of unionized employees and incur \$8,000+ of retirement benefits per employee annually ([Exhibit 37](#)).

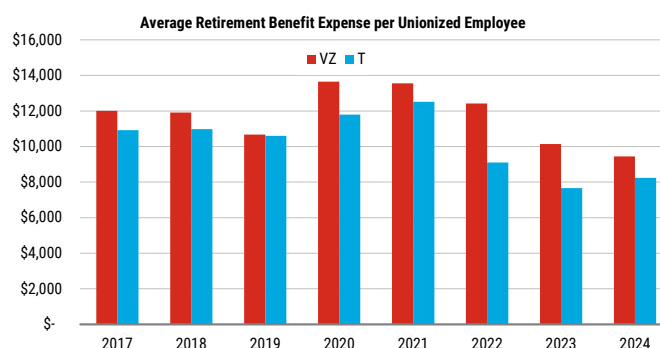
Exhibit 36: Verizon and AT&T have large unionized workforces, with union membership remaining relatively consistent despite workforce reductions



Source: Company filings, Morgan Stanley Research

AT&T and Verizon have achieved a 90%+ funded status on their pension obligations. In recent years, both companies also made pension risk transfers to third parties. In 2023, AT&T transferred \$8 billion of pension obligations and plan assets to Athene, and in 2024, Verizon transferred \$6 billion of pension obligations and plan assets to Prudential and RGA Reinsurance. After these deals, AT&T's remaining pension obligation is \$31 billion, and Verizon's benefit obligation is \$8 billion. In 2024, the unfunded portion of these obligations stood at \$3 billion for AT&T and \$1.1 billion for Verizon ([Exhibit 38](#)).

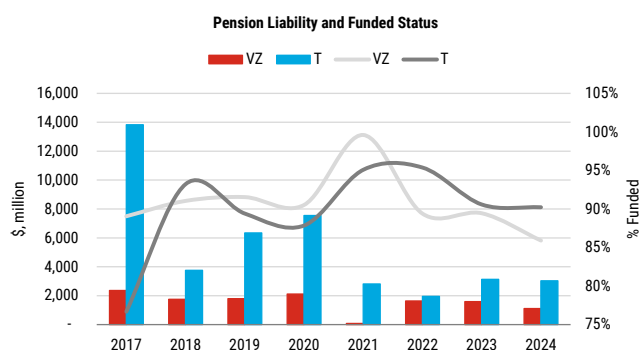
Exhibit 37: Verizon and AT&T incur \$8k-\$10k of retirement benefit expense per unionized employee



Source: Company filings, Morgan Stanley Research

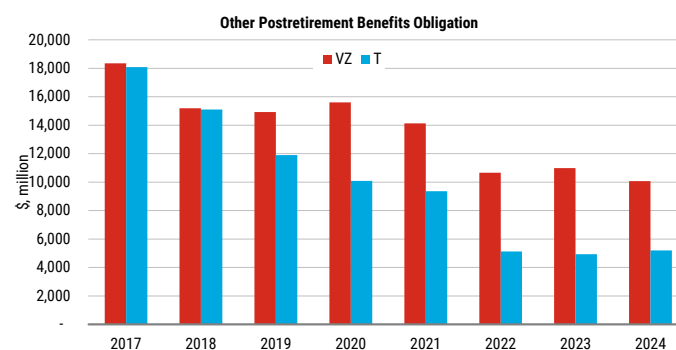
Neither company funds their other postretirement benefit obligations, but AT&T has been curtailing these benefits, which has reduced its liability substantially. In addition, both liabilities decreased in 2022 due to higher interest rates used to discount the liability. See [Exhibit 39](#) below for the OPEB liability decline, which has stabilized in the last three years.

Exhibit 38: Verizon and AT&T have achieved 90% funded status on their pensions, reducing the unfunded pension obligation



Source: Company filings, Morgan Stanley Research.

Exhibit 39: Verizon and AT&T have seen declining OPEB liabilities due to higher discount rates and AT&T's cuts to OPEB benefits



Source: Company filings, Morgan Stanley Research.

We include the unfunded portion of pension and other postretirement benefit obligations in our enterprise value calculation. Because retirement benefits are tax deductible expenses, we tax-adjust the liability, reducing it by the corporate tax rate.

Exhibit 40: We include VZ's pension and OPEB liabilities in our EV calculation. We reduce the \$10 billion obligation by the company's statutory tax rate.

	Current Price		
	2024	2025E	2026E
(\$ Millions, except per-share data)			
Fully Diluted Shares Outstanding	4,214.70	4,223.62	4,226.04
Share Price	\$43.19	\$43.19	\$43.19
Total Market Value	\$182,032.9	\$182,418.0	\$182,522.7
Plus:			
Debt	\$144,014.0	\$140,280.3	\$137,251.5
Other Liabilities	0.0	0.0	0.0
Total Debt	\$144,014.0	\$140,280.3	\$137,251.5
Preferred Stock	\$0.0	\$0.0	\$0.0
Minority Interest	\$1,338.0	\$1,338.0	\$1,338.0
Other Obligations (Pension/OPEB)	\$7,747.6	\$6,913.2	\$5,940.4
Less:			
Cash and Cash Equivalents	\$4,194.0	\$4,405.3	\$6,397.0
Enterprise Value	\$330,938.5	\$326,544.2	\$320,655.7
Total EBITDA	\$48,791.0	\$50,341.9	\$51,886.5
EV / EBITDA	6.8x	6.5x	6.2x
Forward Total EBITDA	\$50,341.9	\$51,886.5	\$52,987.7
EV / Fwd. EBITDA	6.6x	6.3x	6.1x

Source: Company Filings & Morgan Stanley Research

Exhibit 41: We include AT&T's pension and OPEB liabilities in our EV calculation. We reduce the \$8.7 billion liability by the corporate tax rate.

	Current Price		
	2024	2025E	2026E
Shares Outstanding (FD)			
	7,181.00	7,112.62	6,874.40
Share Price	\$27.16	\$27.16	\$27.16
Total Market Value	\$195,036.0	\$193,178.7	\$186,708.6
Plus:			
Total Debt	\$123,532.0	\$119,873.4	\$114,538.9
Preferred Stock	\$20,280.0	\$20,280.0	\$20,280.0
Minority Interest	\$0.0	\$0.0	\$0.0
Other Obligations (Pension/OPEB)	\$7,002.6	\$7,241.2	\$7,489.6
Less:			
Cash and Cash Equivalents	\$3,298.0	\$8,592.6	\$3,892.0
Enterprise Value	\$342,552.6	\$331,980.8	\$325,125.1
Total EBITDA	\$44,759.2	\$46,376.7	\$47,978.0
EV / EBITDA	7.7x	7.2x	6.8x
Forward Total EBITDA	\$46,376.7	\$47,978.0	\$50,195.3
EV / Fwd. EBITDA	7.4x	6.9x	6.5x

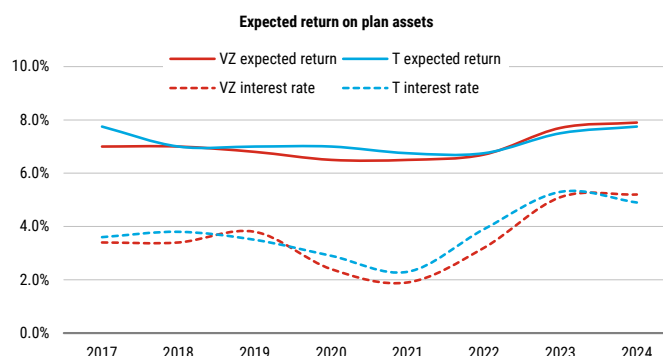
Source: Company Filings & Morgan Stanley Research

AT&T and Verizon elect to recognize actuarial gains and losses in income in the period in which they occur and then adjust non-GAAP earnings for these gains and losses. These gains and losses are often very lumpy, so investors would not want to treat them as recurring. After adjusting for actuarial gains and losses, AT&T and Verizon still typically have pension income remaining due to a higher expected return on plan assets than the discount rate used to measure the pension obligation ([Exhibit 42](#)). In addition, the benefit from AT&T's plan curtailments are amortized through income over time, and these gains are contributing to both GAAP and non-GAAP income. Pension income makes up more than 10% of AT&T's non-GAAP pretax income ([Exhibit 43](#)), which many investors would not consider core operating earnings.

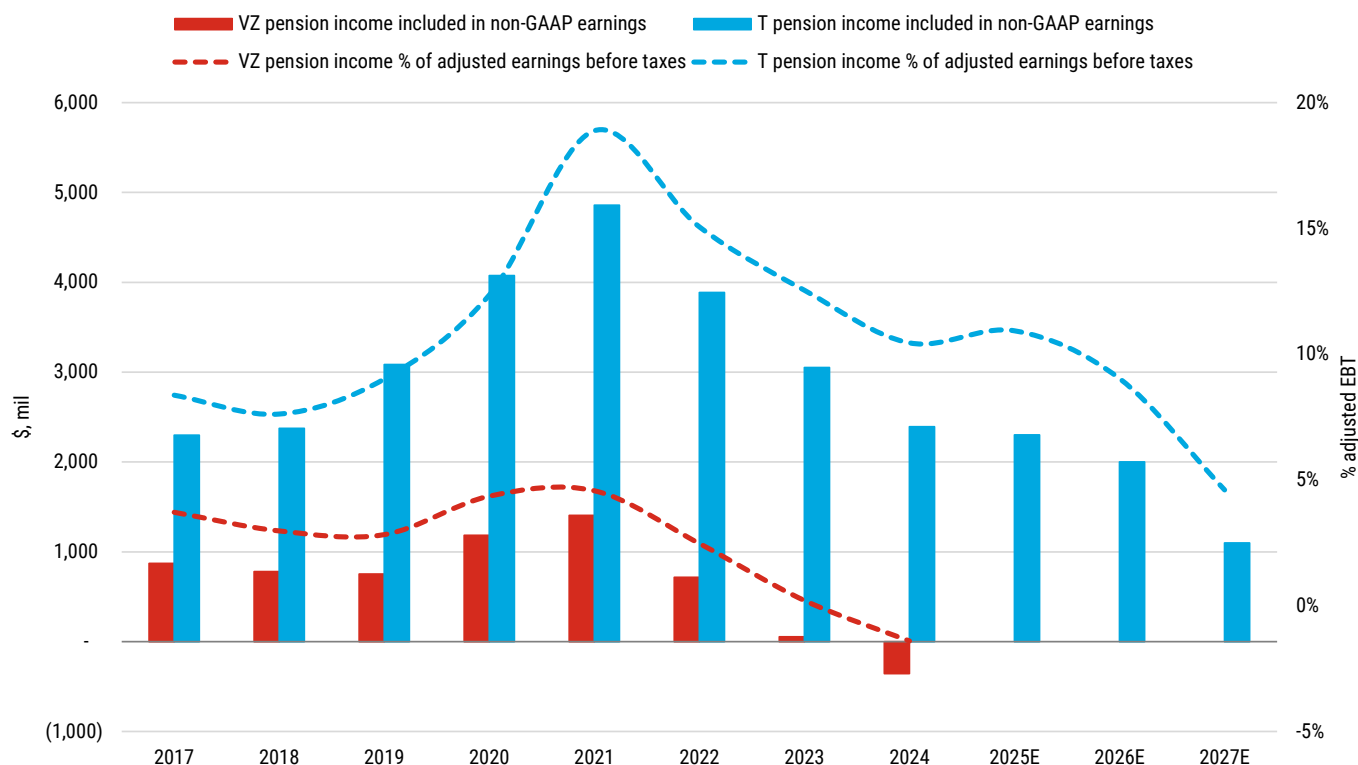
We exclude forecasted pension earnings from our adjusted EPS for valuation purposes. Pension income is non-cash, non-operating, and non-recurring, so we do not put a multiple on these earnings. We do forecast contributions to pension funds and payments for other postretirement benefits that directly impact operating cash flows.

Exhibit 43: AT&T, and Verizon to a lesser extent, have had substantial pension income in their non-GAAP earnings. AT&T continues to guide to pension income for the next three years, which we exclude from our earnings forecast.

Exhibit 42: AT&T and VZ generate pension income because their pension assets generate a higher return than the interest cost on the liability



Source: Company filings, Morgan Stanley Research.



Source: Company filings & Morgan Stanley Research, earnings before taxes adjusted for management's non-GAAP adjustments.

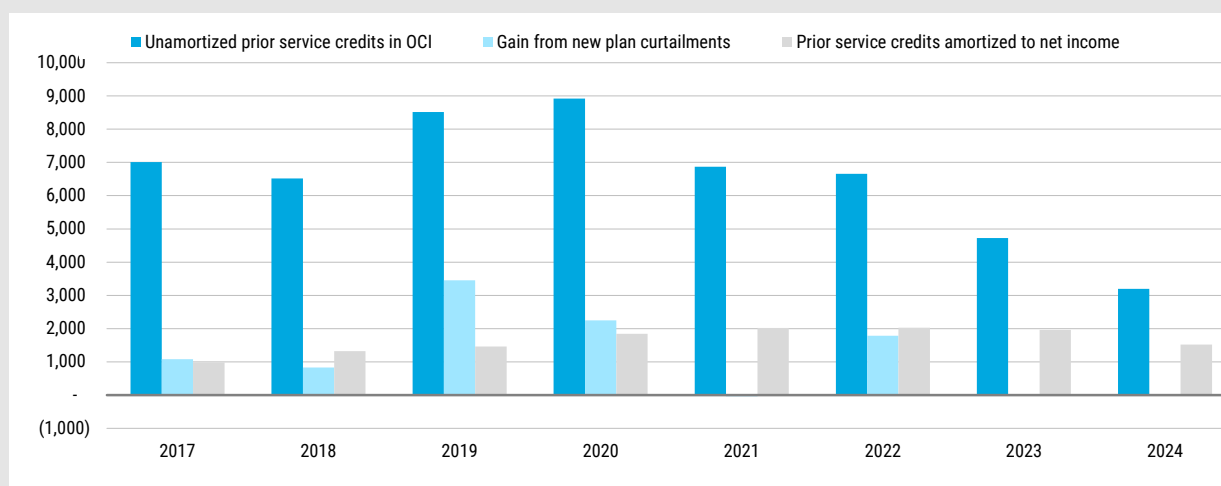
Investor Insight: AT&T's pension and postretirement benefit curtailment gains are initially recorded in Other Comprehensive Income (outside of the P&L) and amortized through income over the employees' estimated remaining service period. Unless further plan curtailments are implemented, these gains will eventually be depleted. From 2021-2023, AT&T recognized ~\$2 billion of income annually from plan curtailments ([Exhibit 45](#)). The company guided to \$1.9 billion of income from amortization of these credits for 2025, \$1.6 billion for 2026, \$0.7 billion for 2027, and less than \$0.1 billion for 2028. We exclude these gains because they are non-recurring.

Exhibit 44: AT&T guides for non-cash pension income to gradually decline over the next 3 years and end in 2028 in the last footnote to the quarterly Financial and Operational Trends update

¹² Other Income (Expense) – Net includes certain items, which have been excluded from adjusted EPS. These adjustments are reflected in the following line items of the schedule: (1) Actuarial and settlement gain/(loss), net; (2) Benefit-related gain (loss), impairments of investments and other; and (3) Interest cost that reflects impacts to our 2022 quarterly benefit expense accruals that resulted from quarterly remeasurements of plan assets and obligations, which included increases in the assumed discount rates. See Adjusting Items in the Financial and Operational Schedules & Non-GAAP Reconciliations. Prior service credit amortization, a component of net pension and postretirement credit, amounted to \$2.0 billion in 2024. Estimated pre-tax amortization of prior service credit for the next four years is: \$1.9 billion for 2025, \$1.6 billion for 2026, \$0.7 billion for 2027, and less than \$0.1 billion for 2028.

Source: Company filings & Morgan Stanley Research.

Exhibit 45: AT&T unamortized prior service credits in OCI have been declining due to fewer benefit curtailments in recent years. The benefit to income from amortizing these credits will ultimately run out unless new cuts to retirement benefits occur.



Source: Company filings, Morgan Stanley Research. Note: Amounts in OCI are after-tax.

Taxes

Verizon and AT&T have historically benefitted from domestic tax policies and are arguably some of the best-positioned companies to benefit from upcoming tax legislation. Because these companies spend substantial amounts on domestic infrastructure annually, reinstating bonus depreciation would meaningfully accelerate FCF and free up cash to return to shareholders through dividends or buybacks. Domestic manufacturing policy details are yet to be determined and also a potential tailwind.

Uncertain Tax Benefits Are Actually Potential Cash Liabilities

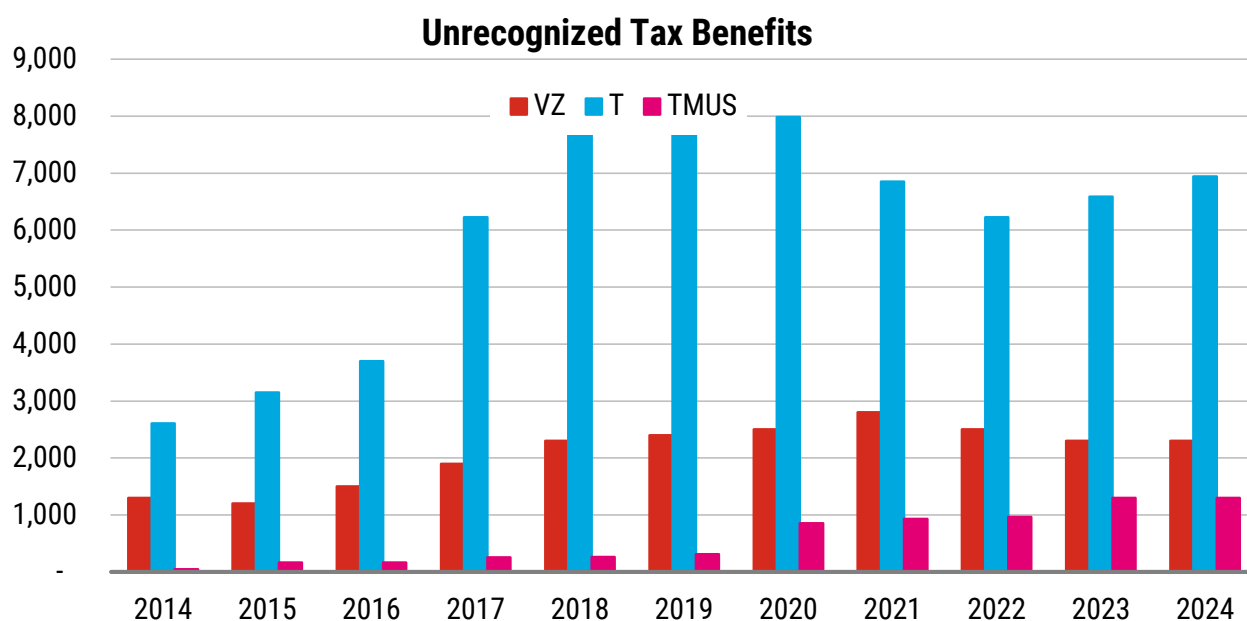
An uncertain tax position is a position taken by a company on its tax return that may not hold up upon examination by tax authorities. In the period that the company takes a tax position that may be challenged in the future by the IRS, the company pays lower income taxes to the IRS, but a tax position can only be recognized in the GAAP financial statements if it is more likely than not that it will be sustained upon review. If the tax position does not meet the more likely than not threshold, typically interpreted as a greater than 50% likelihood, a liability is recognized for the uncertain tax benefit.

The uncertainty around the tax position is ultimately resolved either upon review by the tax authorities or once the audit period expires. If the tax position is not upheld, the company will have to pay the tax with any interest and penalties. If the company had properly

recorded a liability for the uncertain tax benefit (UTB), the reversal would not impact the income in that period. However if the company were under-reserved for the uncertain position, it would increase income tax expense in that period. If the tax position is upheld and the company had recorded a liability for that tax position, the reserve would be released and the income tax benefit would decrease GAAP tax expense in that period.

As seen in [Exhibit 46](#), AT&T has substantial uncertain tax benefits. This indicates that the company likely has a sophisticated tax strategy to take advantage of the investment incentives in the tax code. The UTB liability balance indicates potential future cash tax payments if their strategies are not upheld upon IRS review.

Exhibit 46: Unrecognized tax benefits, net of timing items, are potential cash tax liabilities that companies may owe if the IRS rules against their historical tax positions



Source: Company filings, Morgan Stanley Research.

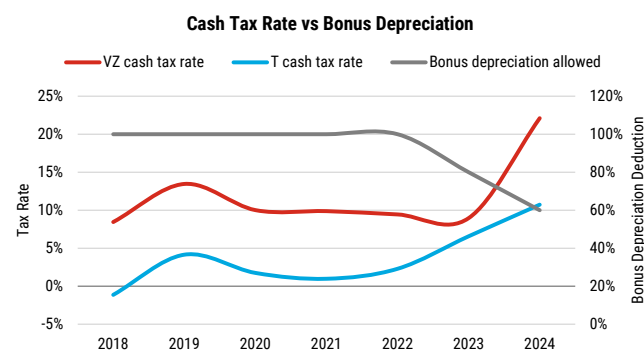
Bonus Depreciation

Bonus depreciation is a tax policy that allows companies to immediately deduct up to the full cost of capital assets in the year of purchase rather than depreciating the cost over the useful life. Under the Tax Cuts and Jobs Act (TCJA), companies were allowed to deduct 100% of non-real estate capex up-front. Starting in 2023, the immediate "bonus" deduction began to phase-out over five years, with an 80% bonus in 2023, 60% in 2024, 40% in 2025, and 20% in 2026.

Ultimately, the value of bonus depreciation is a timing dynamic. Up-front deductions are more valuable than deductions spread over time due to the time value of money - a dollar saved today is worth more than a dollar saved in the future. It is important to note that bonus depreciation is applied on top of an already accelerated Modified Accelerated Cost Recovery System (MACRS), which front loads depreciation deductions.

The example in [Exhibit 48](#) illustrates the economic benefit from 100% bonus depreciation relative to MACRS. For a 10-year capital asset, bonus depreciation reduces the economic cost by as much as 4.5% using a 7% discount rate. The economic benefit from bonus depreciation increases as the depreciable life of the capital asset increases (and decreases for shorter-lived assets).

Exhibit 47: VZ and T cash tax rates have been increasing as bonus depreciation phases out. If bonus depreciation were to be restored to 100%, we think these effective tax rates could reset closer to the pre-phaseout level. We exclude TMUS because NOLs are sufficient to offset current income.



Source: Company Filings and Morgan Stanley Research. EBT adjusted for company non-GAAP adjustments.

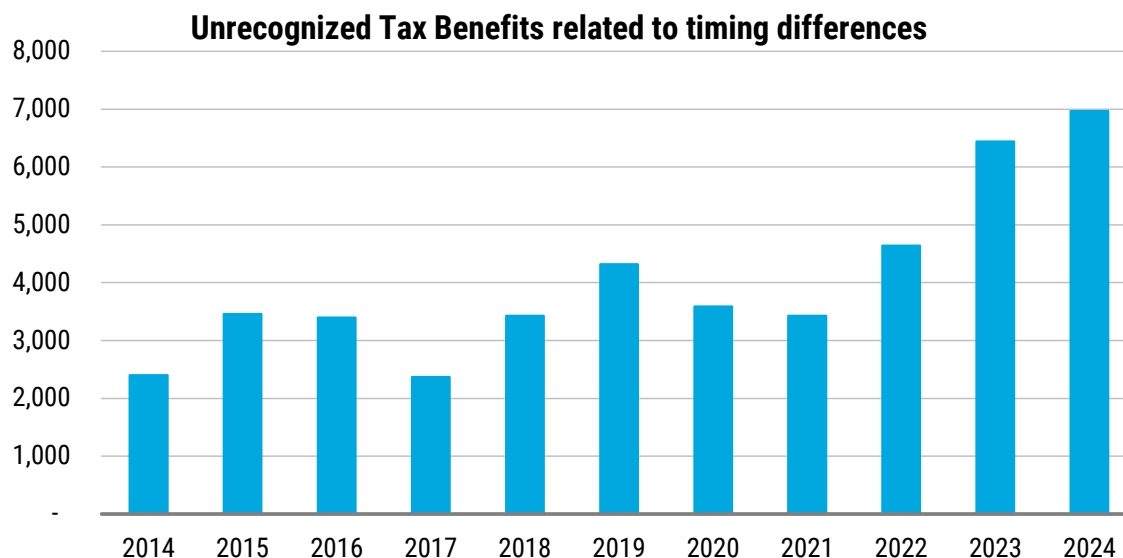
In addition to the unrecognized tax benefits in the chart in [Exhibit 46](#), AT&T has substantial uncertain tax benefits from "timing differences" (VZ and TMUS do not disclose material timing differences). [Exhibit 35](#) shows AT&T has ~\$7 billion of unrecognized tax benefits attributable to "timing differences" at YE 2024, which may be related to tax positions related to bonus depreciation.

Exhibit 48: Illustrative example of bonus depreciation economic impact

10-Year Property	0	1	2	3	4	5	6	7	8	9	10
MACRS Depreciation Factor	10.00%	18.00%	14.40%	11.52%	9.22%	7.37%	6.55%	6.55%	6.56%	6.55%	3.28%
100% Bonus Depreciation	100,000	0	0	0	0	0	0	0	0	0	0
MACRS Depreciation	10,000	18,000	14,400	11,520	9,220	7,370	6,550	6,550	6,560	6,550	3,280
Depr. Variance	90,000	(18,000)	(14,400)	(11,520)	(9,220)	(7,370)	(6,550)	(6,550)	(6,560)	(6,550)	(3,280)
Tax adjusted Variance (Cash Flows)	18,900	(3,780)	(3,024)	(2,419)	(1,936)	(1,548)	(1,376)	(1,376)	(1,378)	(1,376)	(689)
Value of Income Tax Benefit (NPV of tax differential)	4,497										
As % of capital expenditure	4.5%										

Note: assuming a 7% discount rate
Source: Morgan Stanley Research

Exhibit 49: AT&T Unrecognized Tax Benefits attributable to "Timing Items" suggest a historical approach to capitalization that led to significant bonus depreciation benefits - supporting view AT&T may benefit the most from future bonus depreciation changes



Source: Company Filings & Morgan Stanley Research.

Investor Insight: We believe AT&T would likely be the largest beneficiary of reinstating full bonus depreciation tax policy based on their disclosure of significant unrecognized tax benefits attributed to timing differences, which we think could be related to timing of depreciation expense.

On the Q4 2024 earnings call, AT&T said cash taxes excluding DirecTV will increase by \$1.5 billion to \$3.3 billion for 2025, followed by an additional \$1 billion in 2026 "based on the current tax law, including the continued phase out of bonus depreciation." If bonus depreciation were to be reinstated, they could have "more cash flow growth than EBITDA growth potentially."

At the March 2025 TMT conference, Verizon addressed the impact of bonus depreciation on their taxes, saying that the unwinding of bonus depreciation is "going to put pressure on cash taxes." The company followed that up by indicating that "if the provisions come back the way they were in 2017, it could be meaningful for us."

The additional FCF from bonus depreciation savings could be invested back in the business or additional share buybacks.

Amortization of Acquired Intangibles

While many companies exclude amortization of acquired operating intangible assets from their earnings, we include the amortization expense in EPS for valuation purposes to fully reflect the costs associated with the revenues generated by these operating assets.

Intangible assets are most often recorded in an acquisition. At the time of acquisition, an acquirer allocates the purchase price to the target company's assets and liabilities, and records those amounts on its balance sheet. Any remaining purchase price is allocated to goodwill. Intangible assets with limited useful lives such as customer relationships, technology, and other IP are then amortized to net income over their remaining estimated useful lives.

Many companies will exclude the cost of acquired intangibles from their non-GAAP earnings. When a company is acquiring operating assets through acquisitions, we believe the cost of these assets should be included in any measure of earnings for two primary reasons:

1. Most companies have the option to grow organically by investing in existing brands, building out their salesforce to expand the cus-

tomers base, and developing new products, or to make acquisitions to achieve similar goals. In order to compare these two strategies apples-to-apples, we would include amortization of acquired intangibles in earnings.

2. When a company makes an acquisition, it records all of the revenue generated from the acquired intangible assets. If the costs associated with this revenue are excluded from earnings, the company gets all of the revenue (gain) without any associated expense (pain). In order to align revenues and expenses, amortization of acquired intangibles should be included in earnings.

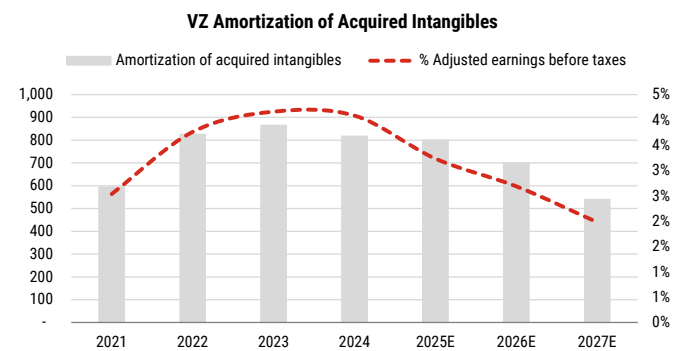
After Verizon's Tracfone acquisition in 2021, the company began adjusting earnings for amortization of acquired intangibles. [Exhibit 50](#) shows Verizon's purchase price allocation for Tracfone. Of the \$7 billion of net assets acquired, \$4.3 billion was allocated to "Other intangible assets." These assets included customer relationships, distribution relationships, trade names, and technology - all operating assets. We include the amortization expense from these intangibles in EPS, and this adjustment is ~4% of Verizon's adjusted pretax earnings ([Exhibit 51](#)).

Exhibit 50: Verizon 2021 10-K: Purchase Price Allocation for Tracfone Acquisition: \$4.3 billion of the purchase price was allocated to operating assets like customer relationships, distribution relationships, trade names, and technology.

	November 23, 2021
(dollars in millions)	
Consideration:	
Cash, net of cash acquired and working capital and other adjustments	\$ 3,491
Fair value of Verizon common stock (57,596,544 shares)	2,981
Fair value of contingent consideration to be paid	542
Total consideration	\$ 7,014
Assets acquired:	
Current assets	\$ 1,370
Property, plant and equipment, net	96
Goodwill	3,723
Other intangible assets	4,374
Other assets	731
Total assets acquired	\$ 10,294
Liabilities assumed:	
Current liabilities	1,433
Deferred income taxes	1,007
Other liabilities	840
Total liabilities assumed	\$ 3,280
Net assets acquired	\$ 7,014

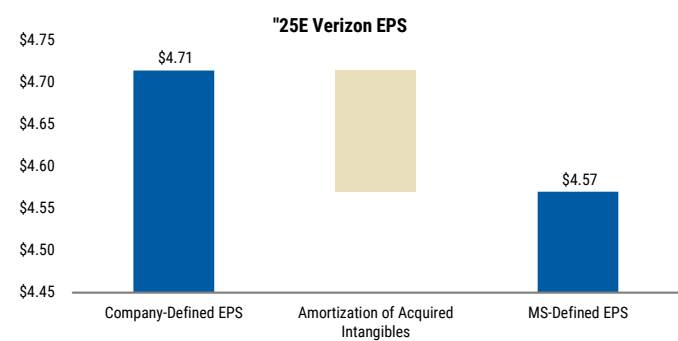
Other intangible assets include \$2.3 billion related to customer relationships, with a weighted-average amortization period of 6 years, \$1.3 billion related to distribution relationships, with a weighted-average amortization period of 5 years, \$744 million related to trade names with a weighted-average amortization period of 16.5 years and \$110 million related to acquired technology, with a weighted-average amortization period of 10 years. The intangible assets were assigned preliminary estimated fair values using an income approach. The valuations are considered Level 3 fair value measurements due to the use of significant inputs not observable in the market, which include the discount rate, royalty rate and amount and timing of future cash flows.

Exhibit 51: Verizon's reported adjusted EPS adds back the amortization of acquired intangibles, which we include in our EPS calculation for valuation purposes and reduces adj. EPS by ~4%. AT&T's amortization of intangibles is largely driven by its soon to be sold DirecTV position, and T-Mobile does not adjust its EPS for this item.



Source: Company Filings and Morgan Stanley Research.

Exhibit 52: We adjust VZ's EPS to include the cost of amortized intangibles. This adjustment does not have a material impact on P/E.



Source: Company Filings and Morgan Stanley Research.

Preferred Interests and Wireless Partnerships

The wireless business has consolidated over the decades, but some partnerships remain in place. These partnerships are typically carried as non-controlling interest or preferred equity.

These are earnings not attributable to common shareholders, typically carry cash distribution obligations, and may have put/call features. We include these minority interests and/or preferred interests as liabilities in building up enterprise value, and we include the expense and/or distributions to partners as reductions to earnings or FCF.

Exhibit 53: We include Preferred Stock and Preferred Interests in our EV calculation for AT&T

	Current Price		
	2024	2025E	2026E
Shares Outstanding (FD)	7,181.00	7,112.62	6,874.40
Share Price	\$27.16	\$27.16	\$27.16
Total Market Value	\$195,036.0	\$193,178.7	\$186,708.6
<i>Plus:</i>			
Total Debt	\$123,532.0	\$119,873.4	\$114,538.9
Preferred Stock	\$20,280.0	\$20,280.0	\$20,280.0
Minority Interest	\$0.0	\$0.0	\$0.0
Other Obligations (Pension/OPEB)	\$7,002.6	\$7,241.2	\$7,489.6
<i>Less:</i>			
Cash and Cash Equivalents	\$3,298.0	\$8,592.6	\$3,892.0
Enterprise Value	\$342,552.6	\$331,980.8	\$325,125.1
Total EBITDA	\$44,759.2	\$46,376.7	\$47,978.0
EV / EBITDA	7.7x	7.2x	6.8x
Forward Total EBITDA	\$46,376.7	\$47,978.0	\$50,195.3
EV / Fwd. EBITDA	7.4x	6.9x	6.5x

Source: Company Filings & Morgan Stanley Research.

AT&T Preferred Stock and Preferred Interests

AT&T uses preferred interests as part of its financing strategy. It issues cumulative perpetual preferred membership interests in certain subsidiaries, including in Tower Holdings (\$6bn in 2019), Telco LLC (\$7.25bn across A-1, A-2, A-3), and Mobility II (\$2bn). By selling redeemable noncontrolling interests in various subsidiaries, AT&T is able to raise capital with more flexibility than issuing debt because it is not required to repay the principal at any certain date. The shares are also non-voting and do not dilute control over the subsidiaries. We consider preferred shares and preferred interests debt-like financing and include them as debt in our enterprise value calculation ([Exhibit 53](#)).

Exhibit 54: AT&T Preferred Stock and Preferred Interests as of 2024

Preferred Stock	2024
Parent Pfd - Series A	1,200
Parent Pfd - Series B (Euro denominated)	2,080
Parent Pfd - Series C	1,750
Tower Holdings Pfd Tranche A	1,500
Tower Holdings Pfd Tranche B	4,500
Mobility II LLC Series B Pfd Membership Interests	2,000
Telco LLC Tranche A	2,000
Telco LLC Tranche B	5,250
Total Preferred Stock	20,280

Source: Company Filings & Morgan Stanley Research.

Verizon Noncontrolling Interests

Verizon has minority interests remaining from previous acquisitions. This means that a portion of the earnings from these subsidiaries does not accumulate to Verizon shareholders. Sometimes companies will make payments to acquire the minority interest, which is another way that cash flow can go to the minority shareholders instead of the parent shareholders. We include NCI as liabilities in our enterprise value build (Exhibit 55).

Exhibit 55: We include minority interest in Verizon's EV calculation

(\$ Millions, except per-share data)	Current Price		
	2024	2025E	2026E
Fully Diluted Shares Outstanding	4,214.70	4,223.62	4,226.04
Share Price	\$43.19	\$43.19	\$43.19
Total Market Value	\$182,032.9	\$182,418.0	\$182,522.7
<i>Plus:</i>			
Debt	\$144,014.0	\$140,280.3	\$137,251.5
Other Liabilities	0.0	0.0	0.0
Total Debt	\$144,014.0	\$140,280.3	\$137,251.5
Preferred Stock	\$0.0	\$0.0	\$0.0
Minority Interest	\$1,338.0	\$1,338.0	\$1,338.0
Other Obligations (Pension/OPEB)	\$7,747.6	\$6,913.2	\$5,940.4
<i>Less:</i>			
Cash and Cash Equivalents	\$4,194.0	\$4,405.3	\$6,397.0
Enterprise Value	\$330,938.5	\$326,544.2	\$320,655.7
Total EBITDA	\$48,791.0	\$50,341.9	\$51,886.5
EV / EBITDA	6.8x	6.5x	6.2x
Forward Total EBITDA	\$50,341.9	\$51,886.5	\$52,987.7
EV / Fwd. EBITDA	6.6x	6.3x	6.1x

Source: Company Filings & Morgan Stanley Research.

Exhibit 56: VZ 2024 10-K: Verizon acquired additional minority interest in 2024

Noncontrolling Interests

During the year ended December 31, 2024, Verizon entered into and completed agreements to acquire additional interests in certain controlled entities for cash consideration of \$280 million. Verizon continues to retain controlling financial interest within these entities; therefore, the changes in ownership interest were accounted for as equity transactions. This resulted in a reduction of additional paid-in capital of \$228 million, reflected in Other, and a reduction of noncontrolling interest of \$52 million, reflected in Distributions and other, both within our consolidated statement of changes in equity for the year ended December 31, 2024. These transactions were recorded within Other, net cash flow from financing activities in our consolidated statement of cash flows for the year ended December 31, 2024.

Source: Company Filings & Morgan Stanley Research.

Stock-Based Compensation

All three of the major US carriers use some SBC to compensate their employees, though to a far lesser degree than seen at tech companies. We include SBC in FCF for valuation purposes to capture the full cost of employee compensation, which reduces our FCF by ~2-4% across the carriers.

Stock-based compensation (SBC) is used to incentivize employee performance, attract and retain talent, and align shareholder and employee interests by compensating employees with company stock instead of cash. The two most common forms of stock-based compensation are restricted stock and stock options. Restricted Stock Units (RSUs) are a promise to give an employee shares of company stock at a future date once certain vesting conditions are met, usually performance-based or time-based vesting. Stock options give the employee the option to purchase company stock at a future date at a set price, so as a company's share price rises, the option becomes more valuable.

SBC is typically settled by the employee receiving shares of company stock, but sometimes companies will choose to settle these awards by paying the employee with cash in an amount equal to the value of the share-based award at the time of vesting. The accounting for SBC is very different for equity settled awards than for cash settled awards.

Accounting for Equity-Settled Stock-Based Compensation

Stock-based compensation expense is initially valued at the fair value on the date it is granted to the employee, and the amount expensed over the vesting period does not change from the initial fair value. In other words, SBC expense is not affected by subsequent changes in the company's stock price. The underlying concept is that the fair value of the SBC at the time it is granted represents the fair value of the employee's services at that point in time, as agreed upon by the employee and the company.

Like other forms of compensation, GAAP requires stock-based plans to be recognized as an expense over the period in which the services are performed or vesting conditions are achieved, so the expense is properly captured in net income.

In the statement of cash flows, SBC is considered a non-cash expense and added back to net income to arrive at cash from operations. This cash flow treatment (no impact on operating cash flow) does not capture the economics the same way that two separate transactions - issuing equity to raise cash (financing inflow), and then paying the employee with that cash (operating outflow) - would be reflected in the financials. Because SBC is considered a non-cash transaction in the cash flow statement, common free cash flow calculations do not include this economic cost.

Companies often use their share buyback authorizations to offset the dilution from SBC. We suggest comparing the amount of any stock repurchased in a period to SBC to see if a buyback is actually returning capital to investors or merely offsetting SBC dilution.

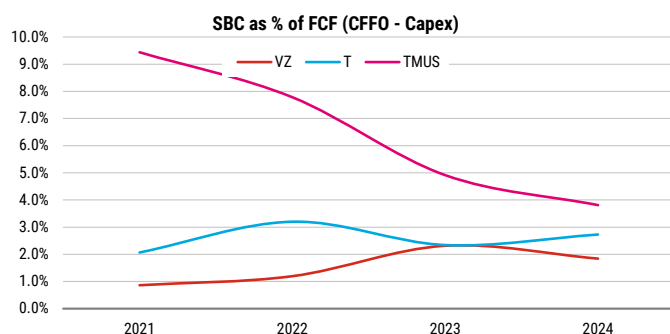
Accounting for Cash-Settled Stock-Based Compensation

If a company settles share-based compensation in cash, the accounting model treats the compensation payable as a liability. That liability is remeasured each reporting period for changes in the stock price to reflect the amount of cash that the company would ultimately have to pay the employee when the award vests. The awards are expensed over the vesting period, consistent with the equity settled awards, and any changes in fair value are also reflected in net income. In the cash flow statement, the cash payments to employees are captured within operating cash flow, so the economic cost is properly captured in free cash flow measures.

Stock-Based Compensation and Free Cash Flow

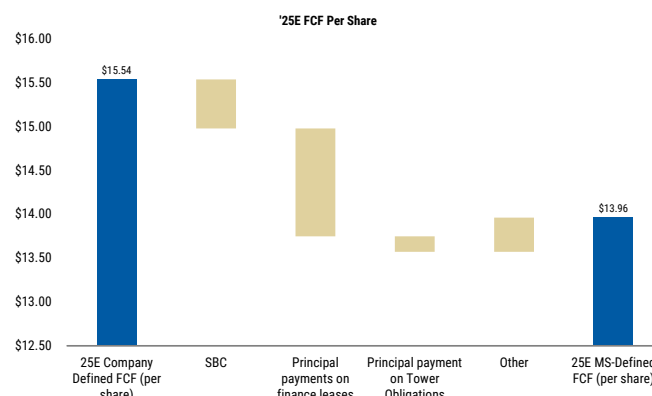
All three of the major US carriers use SBC to compensate their employees. T-Mobile and AT&T use traditional SBC, while half of Verizon's grants are equity-settled and half are cash-settled. We include all SBC in FCF to capture the full employee cost. The SBC adjustment is ~2-3% of FCF for VZ and T, and ~4% at TMUS. [Exhibit 57](#) shows how T-Mobile's use of SBC is converging to a level similar to peers Verizon and AT&T in recent periods, a trend that will likely continue as FCF grows, reducing the impact of our adjustment.

Exhibit 57: T-Mobile's SBC has been converging to levels similar to T and VZ as the business grows



Source: Company Filings & Morgan Stanley Research.

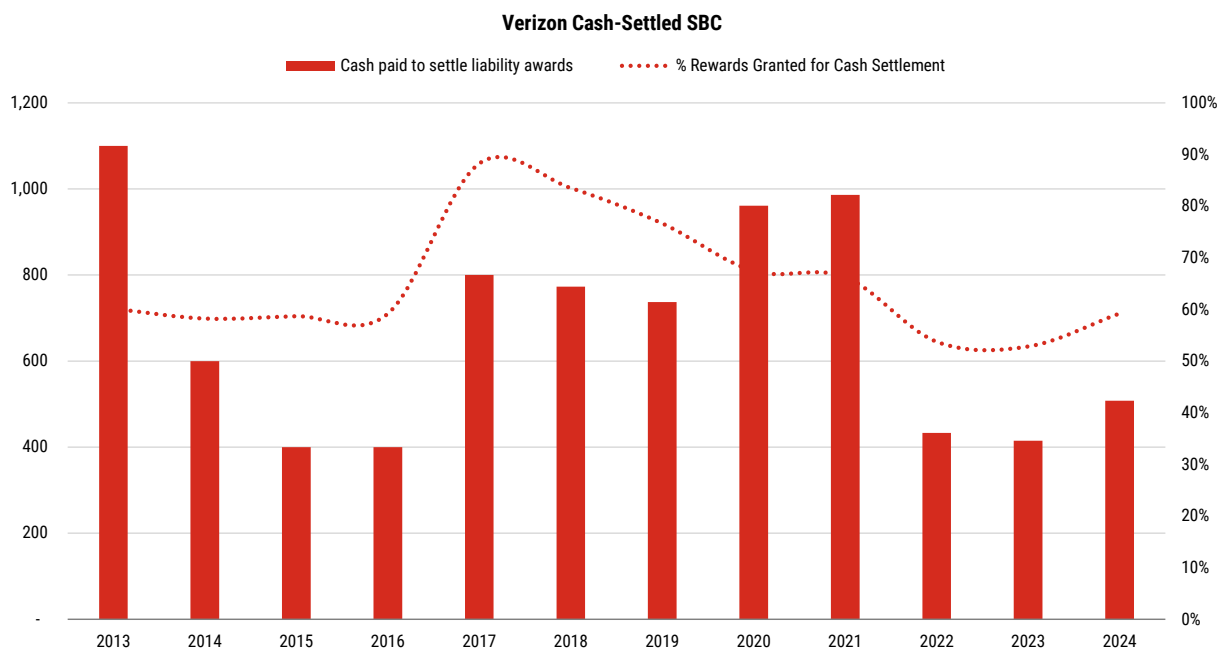
Exhibit 58: T-Mobile has the largest adjustment for stock-based compensation of major carriers. We believe this adjustment will decline relative to reported FCF over time as FCF grows.



Source: Company Filings & Morgan Stanley Research

Verizon has paid as much as \$1 billion for cash-settled SBC, but cash payments have been declining in recent years as the company began granting a more even mix of cash-settled awards and equity-settled awards. [Exhibit 59](#) below shows the evolving trend of cash paid to settle SBC at Verizon.

Exhibit 59: Cash payments to settle SBC has been declining at VZ as cash-settled awards make up less of total grants



Source: Company Filings & Morgan Stanley Research.

Debt Adjustments by Rating Agencies & Credit Investors

Lindsay Tyler, David Hamburger, Kyle Yellin

Alongside this overview of key accounting topics and treatments for US wireless companies, it is worth highlighting that credit investors and ratings agencies sum debt-like liabilities related to these topics (e.g., equipment installment plan receivables, lease liabilities, pension underfunding) when calculating adjusted debt. We introduced this earlier in the note, framing the discussion with four key takeaways, here: [Executive Summary: In Telecom, Accounting Plays A Key Role in Investment Analysis](#).

For the Big 3 US wireless companies, such debt adjustments are sizable, especially compared to the rest of our broader investment-grade TMT fixed-income coverage. See estimated debt adjustments across the Big 3 in [Exhibit 60](#), followed by a detailed table explaining those adjustments by category (focusing on Verizon and AT&T, and Moody's and S&P treatment).

The agencies also make their own EBITDA and FCF adjustments for many of the same items (e.g., adding back operating lease rent), but ultimately the debt adjustments overrun those others adjustments. As a result, adjusted leverage metrics are significantly higher relative to reported ones. In recent history, we've seen ~0.6-0.7 turns of differential between reported net leverage and S&P-adjusted net leverage for the Big 3 US wireless companies, largely driven by the debt adjustments (shown earlier in [Exhibit 11](#)).

For a single company, the agencies and investors alike can have different approaches for calculating debt adjustments; this can be as simple as S&P and Moody's approaching cash differently. And further, comparing those between the Big 3 can illuminate the different approaches management teams have taken (ex. AT&T's differentiated use of preferred shares and preferreds in subsidiaries) and how those have changed over the years, resulting from funding needs, balance sheet size, corporate moves, preferences, and more.

Exhibit 60: We agree with the rating agencies that the Big 3 US wireless names require numerous debt adjustments on top of their reported debt metric; here we estimate and reconcile YE24 estimated debt adjustments for each of the Big 3 largely based on Moody's methodology

Summary Debt Adjustments Largely Aligning with Moody's Approach (\$bn)	AT&T			TMUS			VZ		
	YE20	3/31/24	YE24 Est.	YE20	YE23	YE24 Est.	YE20	6/30/24	YE24 Est.
ST & LT Debt Excl. ABS	157.2	132.8	123.5	71.1	74.3	76.7	118.5	125.3	117.9
(+) Handset ABS	0.0	0.0	0.0	0.0	0.7	1.6	10.6	24.0	26.1
(+) Tower Obligations on Balance Sheet	0.0	0.0	0.0	3.0	3.8	3.7	0.0	0.0	0.0
Approx. Starting Reported Debt per Moody's	157.2	132.8	123.5	74.2	78.8	81.9	129.1	149.3	144.0
(+) Other Securitizations	10.9	9.5	8.2	1.7	1.6	1.5	0.0	0.0	0.0
(+) Pension Underfunding / Obligation	7.5	3.1	3.0	2.3	1.6	0.9	2.1	1.6	1.1
(+) Operating Leases	25.7	21.1	20.9	33.1	34.3	32.0	21.5	23.7	24.3
(+) Preferred Equity at Parent (50%)	2.6	2.6	2.6	0.0	0.0	0.0	0.0	0.0	0.0
(+) Preferred Equity at Subsidiaries	16.0	15.3	15.3	0.0	0.0	0.0	0.0	0.0	0.0
Subtotal of Other Adjustments	62.8	51.7	50.0	37.1	37.5	34.4	23.6	25.3	25.5
Adj. Gross Debt (Estimate for Moody's)	220.0	184.4	173.6	111.3	116.3	116.3	152.7	174.6	169.5
(-) Cash	(9.7)	(3.5)	(3.3)	(10.4)	(5.1)	(5.4)	(22.2)	(2.4)	(4.2)
Adj. Net Debt	210.3	180.9	170.3	100.9	111.1	110.9	130.5	172.1	165.3
All Adjustments (Subtotal above + ABS + Tower Ob.)	62.8	51.7	50.0	40.1	42.0	39.6	34.2	49.3	51.6
Adj. Gross Debt (Estimate for Moody's, from above)	220.0	184.4	173.6	111.3	116.3	116.3	152.7	174.6	169.5
All Adjustments / Adjusted Gross Debt	29%	28%	29%	36%	36%	34%	22%	28%	30%

Source: Company filings, Moody's (Peer Snapshot for AT&T dated 3/27/25; Credit Opinion notes dated 6/5/24 and 1/6/22 for AT&T, 5/8/24 and 2/28/23 for TMUS, 9/10/24 and 6/30/23 for VZ).

Typical Debt Adjustments for US Wireless Companies

STANDARD ADJUSTMENT	ADDITIONAL COMMENTARY & HIGHLIGHTS REGARDING RATING AGENCY TREATMENT
Pension underfunding	• These adjustments are common across most corporate issuers and can be updated annually with the 10-K disclosures. The obligations are treated as debt-like since they represent financial obligations that need to be paid over time and are regulatory/contractual in nature. Read more in Pensions and Other Post-Employment Benefits. • Moody's typically adds as debt the amount by which defined-benefit pension plans are underfunded. • S&P typically adds the tax-affected unfunded portion of pension and other post-employment benefits.
Lease liabilities	• As discussed elsewhere in this note (Finance Leases), the wireless industry is capital intensive, and it makes sense to us that lease obligations are included in adjusted debt balances for these names. • Moody's and S&P generally incorporate both operating leases and finance leases in their adjusted debt balances. • Operating leases are sizable for the US wireless companies.
Tower obligations	TMUS has ~\$3.6bn of tower obligations (at end-1Q25) that it lists as a separate line item on the balance sheet (thereby updated quarterly) and does not include it in reported net debt or leverage. Moody's and S&P add the full amount as an adjustment to debt, as they would leases.
EIP receivables/captive finance	• The Big 3 US wireless companies use securitizations to finance handset purchases. There is a significant working capital investment that they need to make to buy the phones upfront so they offset the front-end cash expense by using receivables securitizations. The three all approach EIP receivables financing a bit differently, as we discussed more from the accounting perspective in Equipment Installment Plan Receivables. Moody's generally takes a different approach to debt treatment than the other two agencies, S&P and Fitch, for these programs. • At Moody's, it is our understanding that the agency includes the full amount of VZ's public ABS balance (~\$26bn at end-1Q25) in its starting debt balance (before adding pension and op. lease adjustments). For AT&T's program, Moody's adds the cash proceeds received to date from the off-balance-sheet EIP program (~\$9bn for EIP only at end-1Q25) as an adjustment to the reported debt balance. • At S&P, as outlined in its April 2025 Credit FAQ, the EIPs qualify as captive finance operations under its criteria, which is reflected in the debt adjustments for both VZ and AT&T. The objective is to analyze the 'core' after deconsolidating the captive financing, since captive entities' assets can typically support higher leverage than industrial assets (as seen in other industries like hardware with fincos). S&P estimates the captive's debt based on its on- and off-balance-sheet EIP receivables divided by an implied debt-to-equity ratio, and then deconsolidates the captive debt (and EBITDA) for adjusted leverage. It is our understanding that, for Verizon, S&P excludes the \$20bn+ of ABS from the adjusted debt calculation, and for AT&T, S&P nets out \$9bn of captive finco adjustment (3x ratio) but adds back off-balance-sheet EIP receivables net cash proceeds to date.
Supplier finance	• S&P regards the securitization, sale, and factoring of trade receivables and other assets generated on an ongoing basis as akin to secured financing and makes an adjustment because it does "not presume the company will have permanent access to the securitization or factoring market and may need to incur conventional debt to replace this source of financing" ("Corporate Methodology: Ratios And Adjustments" 4/1/19). • For example, S&P includes trade receivables sold and direct supplier/vendor financing payables as a debt adjustment at AT&T. • See Supplier Financing Programs and this prior GVAT note.
Hybrid instruments	• The agencies assess some instruments/securities, like preferred stock and subordinated debt, using equity content classification (i.e., basket treatment criteria to assign equity credit). • For example, at YE24 and for some time prior, AT&T has had ~\$5.1bn of cumulative perpetual preferred shares across Series A, Series B, and Series C, which all achieved 50% hybrid equity credit treatment. ~\$2.6bn has been included in Moody's adjusted debt balance (6/5/24 note) and has similarly been included in S&P's too. During 1Q25, AT&T issued a call notice for the Series B and redeemed the 20k shares for €2bn.

Preferred interests in subsidiaries

- AT&T uses preferred interests as part of its financing strategy and has issued cumulative perpetual preferred membership interests in certain subsidiaries. The interests outstanding as of end-1Q25 include 1) \$6bn issued out of Tower Holdings (holds various tower assets) in 2019, 2) \$2bn issued by Mobility II in 2023, and 3) \$9.5bn out of Telco LLC across A-1, A-2, A-3, and recently A-4 in 1Q25.
- From the perspective of the rating agencies, these don't meet the criteria for receiving any equity credit. Further, these interests are all structurally senior to debt at AT&T Inc. Our Accounting team colleagues discuss these from their perspective in [Preferred Interests and Wireless Partnerships](#).
- Using AT&T as the example, Moody's characterizes all as preferred equity at subsidiaries. In addition to the \$13bn (at YE24) related to Tower and Telco LLC, S&P breaks out the remaining \$2bn related to the Mobility II redeemable noncontrolling interest into its 'put right/options/nonredeemable interests' definition. AT&T had issued 2mm Series B Cumulative Perpetual Preferred Membership Interests in Mobility II LLC in June 2023, and a holder of the interests may put the interests at points in the future, while Mobility II may redeem at points in the future (see [T 10-K pg. 85](#)).

Unamortized debt discounts

For these companies there can be a sizable difference between balance sheet debt and the principal amount of debt, so there can be a related addback by the agencies.

Cash (-)

- S&P's leverage ratio is on a net basis, typically netting accessible cash and liquid investments. This is common across most issuers/sectors and the adjustment is typically balance sheet cash, but sometimes the adjustment can differ by company given the assessment of quickly accessible holdings and investments.
- Moody's looks at leverage on a gross basis and does not make this adjustment.

Proportionate consolidation related to stake in asset, for structural subordination purposes

AT&T sold a [30% economic stake in DIRECTV to TPG in 2021](#), deconsolidating results. Since then, Moody's has flagged the proportionate 70% of DIRECTV's debt (~\$5bn) when assessing structural subordination in the debt complex. 70% of DTV's debt was \$5.4bn at 3/31/24 (from 6/5/24 note). However, Moody's does not capture this in its adjusted debt balance that is found in Peer Snapshot and other comparative notes. Further, AT&T recently reached an agreement to sell its 70% stake to TPG and expects the transaction to close in 2H25 (see [9/30/24 announcement](#)), so this will fall away if/when the deal closes.

Key sources & recommended related reading: S&P note dated 4/14/25 "Credit FAQ: Calculating Leverage For Selected U.S. Telecommunications And Cable Companies (2025 Update)"; Moody's Cross-Sector Rating Methodology note dated 10/9/24 "Financial Statement Adjustments in the Analysis of Nonfinancial Corporations".

Credit Valuation Methodology and Risks

Lindsay Tyler

AT&T & TMUS Trade Valuation Methodology and Risks					
TRADE	ENTRY LEVEL	ENTRY DATE	CURRENT LEVEL	RATIONALE	RISKS
Buy TMUS 2.05 2/15/28	+114bp	7/17/23	+68bp	• We do not see substantial spread-tightening potential, but the name continues to rank a bit higher on our Composite Score than T within our framework • We like TMUS's leverage stability, strong FCF after dividends/debt, and FWA expansion	• Cautious sentiment around technicals given potential greater funding needs for share buybacks and investments • Wireless competition impacting pricing and subscriber growth
Buy TMUS 2.55 2/15/31	+135bp	9/18/23	+105bp		
Buy TMUS 2¼ 11/15/31	+137bp	7/17/23	+103bp		
Buy TMUS 2.7 3/15/32	+101bp	1/18/24	+76bp		
Buy S 6¾ 11/15/28	124¾ (2.9%)	1/14/22	+99bp	• Has IG ratings from all three agencies • We think the bonds could see further spread compression to TMUS bullets	• Junior to other notes in structure due to subsidiary guarantee differences • High dollar relative to TMUS bullets
Buy S 8¾ 3/15/32	146½ (3.3%)	1/14/22	+101bp		
Buy SPRNTS 5.152 3/20/28	112¾ (+225bp)	6/3/20	100.6 (4.9%)	• The bonds benefit from collateral asset coverage that increases as debt amortizes/matures • Spread pickup vs. TMUS benchmarks • A1 rating at Moody's, and amortization ongoing	• Less liquid than other parts of TMUS complex
Buy T 3¾ 3/15/53	+101bp	12/16/24	+116bp	• Have become more comfortable with AT&T credit, as reflected by its Composite Score increasing and approaching TMUS's • Appreciate the reasonable recently communicated approach to balancing fiber investments with shareholder returns • Appreciate management communication regarding strategic plans • We view the telecom space as relatively defensive	• Lower growth profile relative to TMUS • Has not yet reached its leverage target • Expect a longer-term negative FCF impact related to the DTV stake sale

IG Canada Comms Trade Valuation Methodology and Risks

TRADE	ENTRY LEVEL	ENTRY DATE	CURRENT LEVEL	RATIONALE	RISKS
Sell RCICN 3.8 3/15/32	+124bp	1/18/24	+121bp	• For 32s rec: Don't think spreads price in fundamental risks given 3.7x leverage (guided PF for structured equity investment) is still elevated vs. IG TMT peers, and competition is intensifying • For 29s rec: Expect targeted debt takeout if and when the new structured equity financing closes; these bonds are high coupon, and the 2029 maturity tower is sizable	• For 32s rec: Still relatively strong business profile given position in Canada comms. market, risk of sizable targeted debt paydown across curve • For 29s rec: Fundamental risks related to rising competition, still-elevated leverage, any potential debt takeout may target other CUSIPs
Buy RCICN 5 2/15/29	+79bp	10/30/24	+96bp		
Sell BCECN 5.2 2/15/34	+108bp	1/13/25	+97bp	• Years of leverage uptick, investment, and dividend growth are now being met with intensifying competition and other pressures, constraining overall near-term growth • Contend 10Y should trade at a discount to BBB index; think current premium to COXENT, FOXA, RCICN is unwarranted	• Stronger-than-expected financial and subscriber performance • Liquidity boost in form of further asset monetization (after MLSE stake and Northwestel sales), dividend cut, or something similar
Sell BCECN 5.55 2/15/54	+129bp		+132bp		

David Hamburger

HY Satellite Trade Recommendations

TRADE	ENTRY LEVEL	ENTRY DATE	CURRENT LEVEL	RATIONALE	RISKS
Buy SATS Secd 6¾ '30	91⅞ (8.4%)	12/4/24	93½ (8.7%)	- We think these notes are being offered at an attractive entry level considering that our views on spectrum valuation would result in those being fully covered at par ~30% LTV (with a 37.5% 1L cap) based on third-party appraiser valuation of ~\$33.1bn to AWS-3 and AWS-4 spectrum that serve as collateral to these notes	- Current market volatility - EchoStar is likely to opt for PIK for the first four coupon payments - Wireless business is still far from profitability
Buy SATS Secd 10¾ '29	104¼ (9.3%)	4/8/25	106¼ (8.7%)	- We like the current entry level vs. where these notes have been trading since issuance. ~30% LTV (with a 37.5% 1L cap) based on third-party appraiser valuation of ~\$33.1bn to AWS-3 and AWS-4 spectrum that serve as collateral to these notes	- Current market volatility - Wireless business is still far from profitability
Buy SATS Secd 5¼ '26	91¾ (10.8%)	12/4/24	94¼ (10.2%)	We expect these notes to be refinanced with secured debt (we see sufficient unencumbered assets between spectrum and satellites) likely issued outside of Hughes, and think mgmt. could use the refinancing as an opportunity to collapse Hughes into EchoStar	- Weak financial performance and increasing leverage could be challenges to get the refi done at Hughes if mgmt. chooses that path instead of refinancing debt outside of the silo - Increasing competition for both consumer and enterprise segments due to disruptive players such as Starlink and Amazon
Buy SATS Unsecd 6⅞ '26	80¼ (21.4%)		83 (23%)		

HY Cable Trade Recommendations

TRADE	ENTRY LEVEL	ENTRY DATE	CURRENT LEVEL	RATIONALE	RISKS
Buy CSCHLD gtd 5½ '27	88 (11.1%)	10/15/24	93½ (9.2%)	• We think guaranteed notes are fully covered at par under our estimated \$18.5bn EV for CSC Holdings Restricted Group • We think lower-dollar notes provide a lower attachment value to cushion against LME (liability management exercise) that could potentially seek to capture discounts	• Lower likelihood of potential M&A or asset sales given current market conditions • Weak operational and financial performance with weak FCF expected for the restricted group • Ongoing negotiations between creditors and the company could result in volatility • A potential solution for the capital structure might come with negative implications for longer-dated guaranteed notes • With the relevant credit protections associated to only one TL tranche, we think mgmt. might look to take it out to then access \$20bn+ (per our estimates) of restricted investments capacity
Buy CSCHLD gtd 5% '28	84½ (11.1%)		88¾ (10.3%)		
Buy CSCHLD gtd 6½ '29	84 (11.3%)		82¾ (12.5%)		
Buy CSCHLD gtd 4½ '30	73½ (10.0%)		69¾ (11.6%)		
Buy CSCHLD gtd 3% '31	71½ (9.5%)		68 (11%)		
Buy CSCHLD gtd 4½ '31	73½ (9.8%)		69 (11.3%)		
Sell CHTR 5.375% '29 /	+198bp/	12/11/24	+228bp/	• Expect differential to revert back to at least ~100bp+ on intense competition in the broadband industry, higher buybacks and higher net issuance in 2025 vs. the last two years.	• Lower buybacks in case management has to keep leverage on the lower end of its target (4.0-4.25x) • HY bonds continue to outperform IG bonds
Buy CHTR 5.05% '29	+111bp		+136bp		
Sell CHTR 4.5% '32 /	+237bp/	12/11/24	+257bp/		
Buy CHTR 4.4% '33	+140bp		+182bp		

Current levels updated as of 4/30/25.

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(as of April 30, 2025)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1501	40%	377	46%	25%	691	40%
Equal-weight/Hold	1667	44%	375	45%	22%	796	46%
Not-Rated/Hold	3	0%	0	0%	0%	1	0%
Underweight/Sell	603	16%	76	9%	13%	229	13%
Total	3,774		828			1717	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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The following authors are neither Equity Research Analysts/Strategists nor Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity or fixed income securities: Todd Castagno, CFA, CPA; Clinton Chang, CFA, CPA.

The following authors are Fixed Income Research Analysts/Strategists and are not opining on or expressing recommendations on equity securities: Lindsay A Tyler; David M Hamburger; Kyle B Yellin.

INDUSTRY COVERAGE: Telecom & Cable Services

COMPANY (TICKER)	RATING (AS OF)	PRICE* (05/05/2025)
Benjamin Swinburne, CFA		
Altice USA Inc (ATUS.N)	E (12/16/2020)	\$2.53
AT&T, Inc. (T.N)	O (12/16/2024)	\$27.51
BCE Inc. (BCE.TO)	U (12/16/2024)	C\$29.54
Charter Communications Inc. (CHTR.O)	E (12/09/2021)	\$394.24
Comcast Corporation (CMCSA.O)	E (12/16/2024)	\$34.45
EchoStar Corporation (SATS.O)	E (12/16/2024)	\$23.88
Frontier Communications Parent, Inc. (FYBR.O)	++	\$36.54
Rogers Communications, Inc. (RCIb.TO)	U (12/16/2024)	C\$35.10
TELUS Corp. (T.TO)	E (07/16/2021)	C\$20.85
T-Mobile US, Inc. (TMUS.O)	O (04/17/2020)	\$248.30
Verizon Communications (VZ.N)	++	\$43.54
Cameron Mansson-Perrone		
Sirius XM Radio Inc. (SIRI.O)	U (09/24/2024)	\$21.29

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* Historical prices are not split adjusted.

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The Americas

1585 Broadway
New York, NY 10036-8293
United States
Tel: +1 (1) 212 761 4000

Europe

20 Bank Street, Canary Wharf
London E14 4AD
United Kingdom
Tel: +44 (0) 20 7 425 8000

Japan

1-9-7 Otemachi, Chiyoda-ku
Tokyo 100-8104
Japan
Tel: +81 (0) 3 6836 5000

Asia/Pacific

1 Austin Road West
Kowloon
Hong Kong
Tel: +852 2848 5200